

Fourier Transform and Its Applications

Instructor

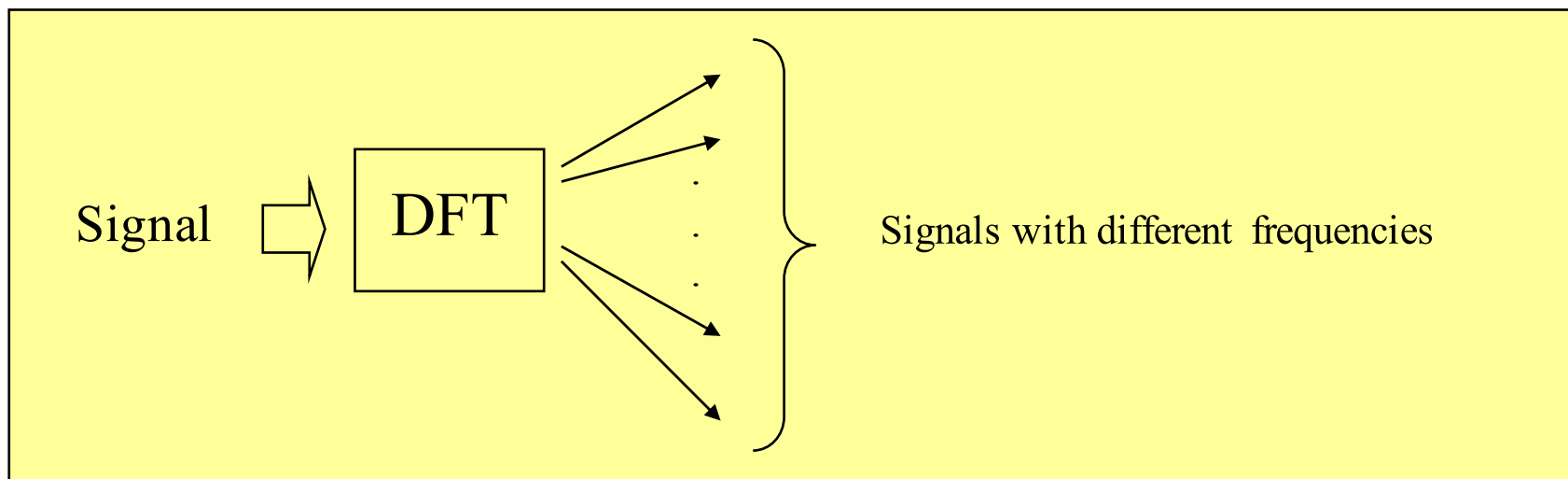
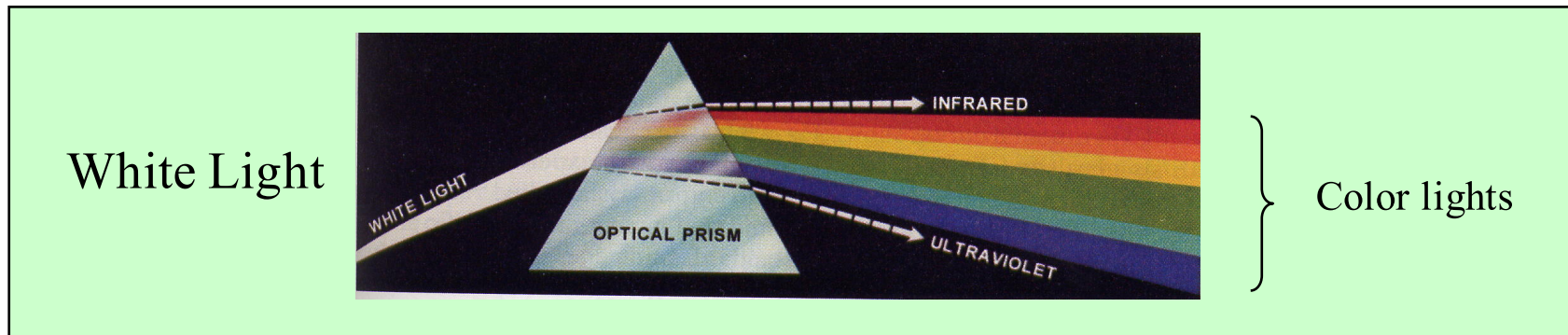
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Outline

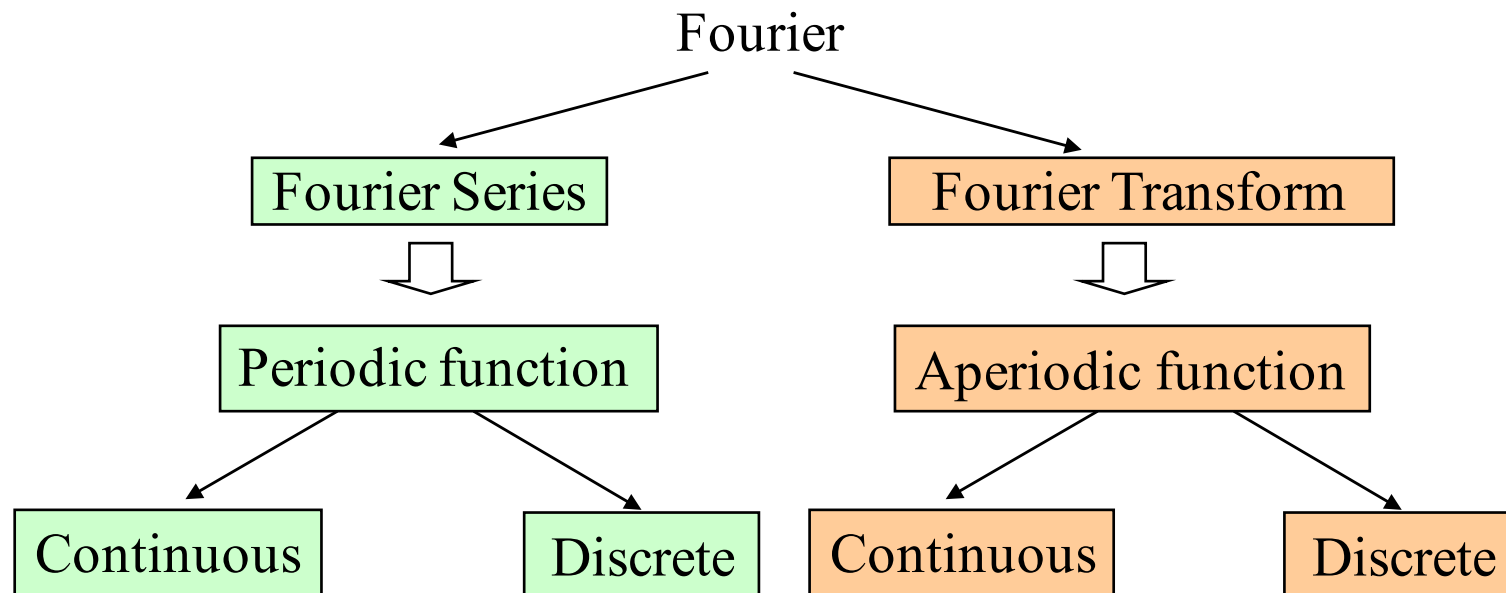
- ❖ Fourier Transform
- ❖ Its Applications

Fourier Analysis



Fourier Theory

$$\text{Any functions or signals} = \sum_{u,v} A \sin(\theta) + B \cos(\theta)$$



1D Discrete Fourier Transform Formula

Forward Transform:

from **space domain** to **frequency domain** (Fourier Domain)

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j\frac{2\pi}{M}ux} \quad u = 0, 1, \dots, M$$

Backward Transform (Inverse Transform):

from **space domain** to **frequency domain** (Fourier Domain)

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j\frac{2\pi}{M}ux} \quad x = 0, 1, \dots, M$$

1D Discrete Fourier Transform

Interpretation

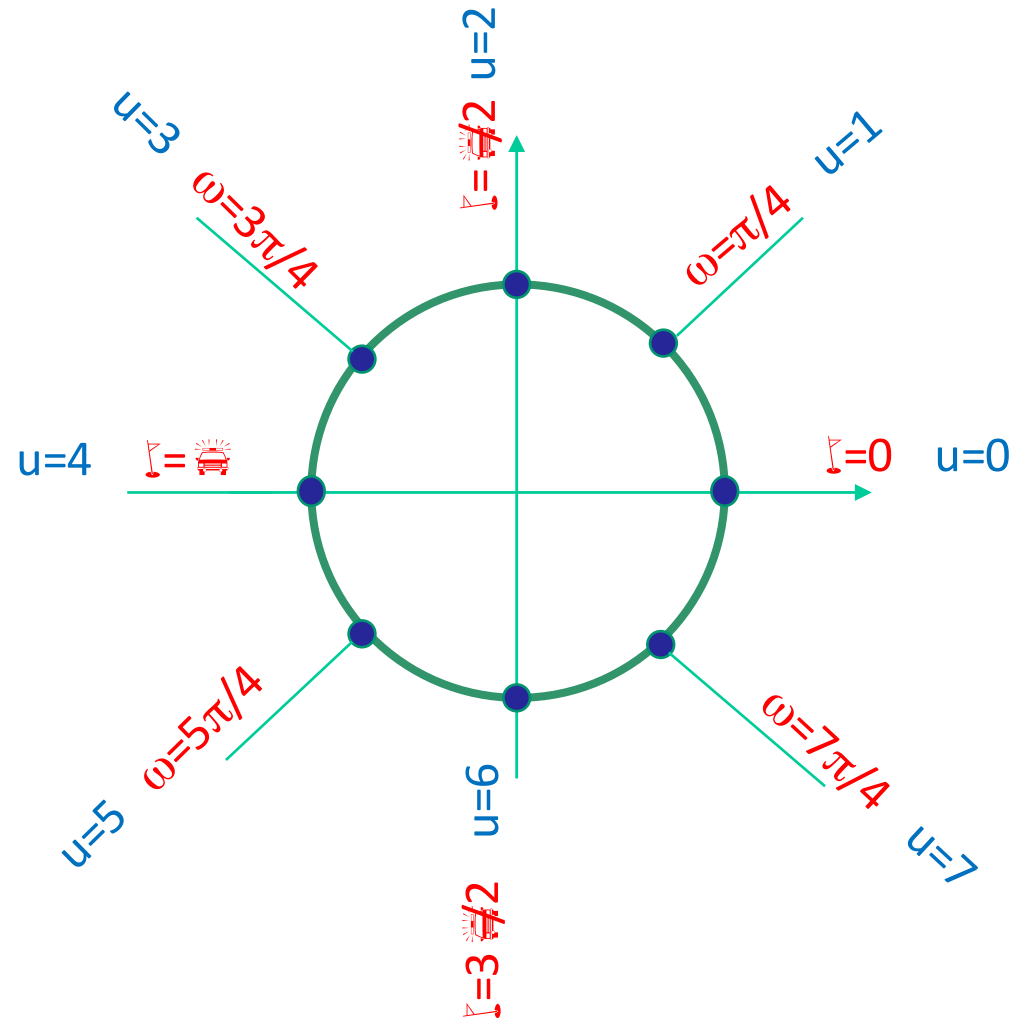
Forward Transform:

$$F(u) = \frac{1}{M} \sum_{x=0}^{M-1} f(x) e^{-j2\pi \left(\frac{ux}{M} \right)}$$

- M : Number of frequencies that we want to decompose $f(x)$ into them
- u : index of decomposed frequencies, $u = 1..M-1$
- For example, $M = 8$, i.e., to decompose into 8 components that their frequencies are $\frac{2\pi}{8}u$, $u = 0..7$

1D Discrete Fourier Transform Interpretation

M frequencies
in a trigonometric circle,
For $M = 8$:



1D Discrete Fourier Transform Interpretation

Fourier transform's meaning:

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j \frac{2\pi}{M} ux} \quad u = 0, 1, \dots, M$$

$F(u)$ tells that how much frequency component $\frac{2\pi}{M}u$ contribute to signal $f(x)$. In the case that $F(u) = 0$ then there is no frequency $\frac{2\pi}{M}u$ the input signal. Otherwise, if the magnitude of $F(u)$ is significantly larger than other frequencies' then frequency $\frac{2\pi}{M}u$ contribute much to the signal, and the shape of the signal tends to be similar with the shape of frequency $\frac{2\pi}{M}u$.

1D Discrete Fourier Transform Interpretation

Fourier transform meaning:

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j \frac{2\pi}{M} ux} \quad u = 0, 1, \dots, M$$

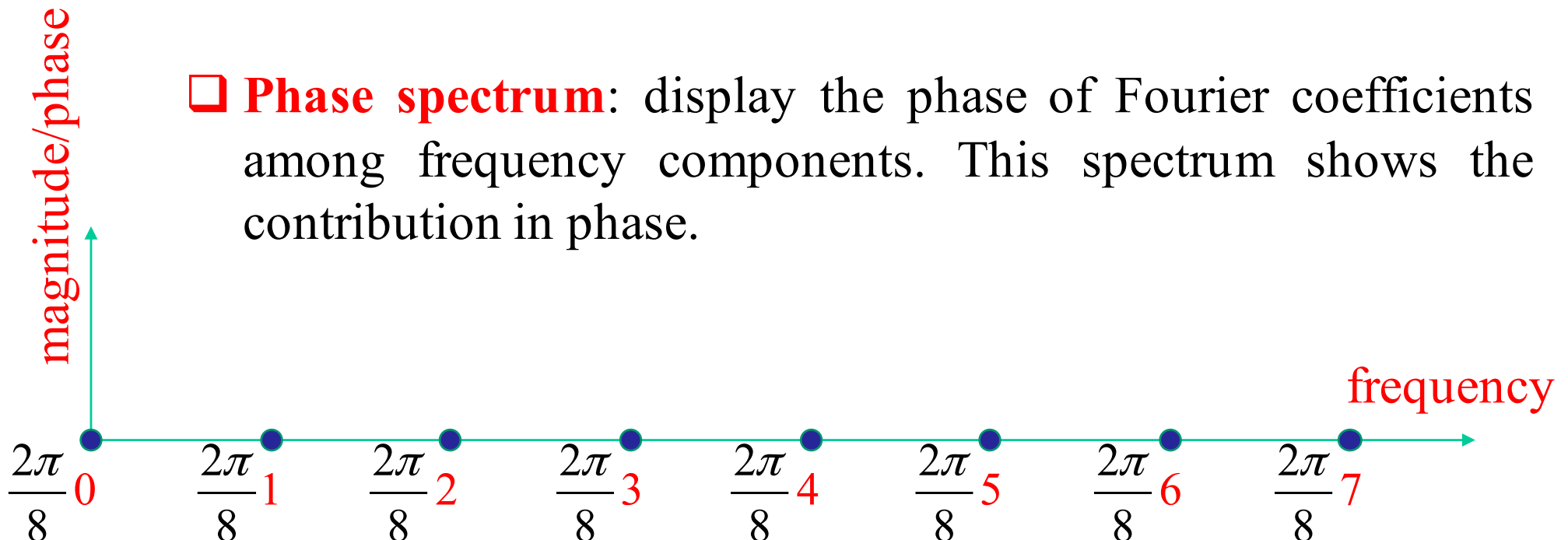
In dot product/ correlation

1D Discrete Fourier Transform Interpretation

Fourier transform's meaning:

□ **Magnitude spectrum:** display the magnitude of Fourier coefficients among frequency components. This spectrum can tell how much a frequency contribute to the input signal.

□ **Phase spectrum:** display the phase of Fourier coefficients among frequency components. This spectrum shows the contribution in phase.



1D Discrete Fourier Transform Interpretation

Fourier transform's meaning:

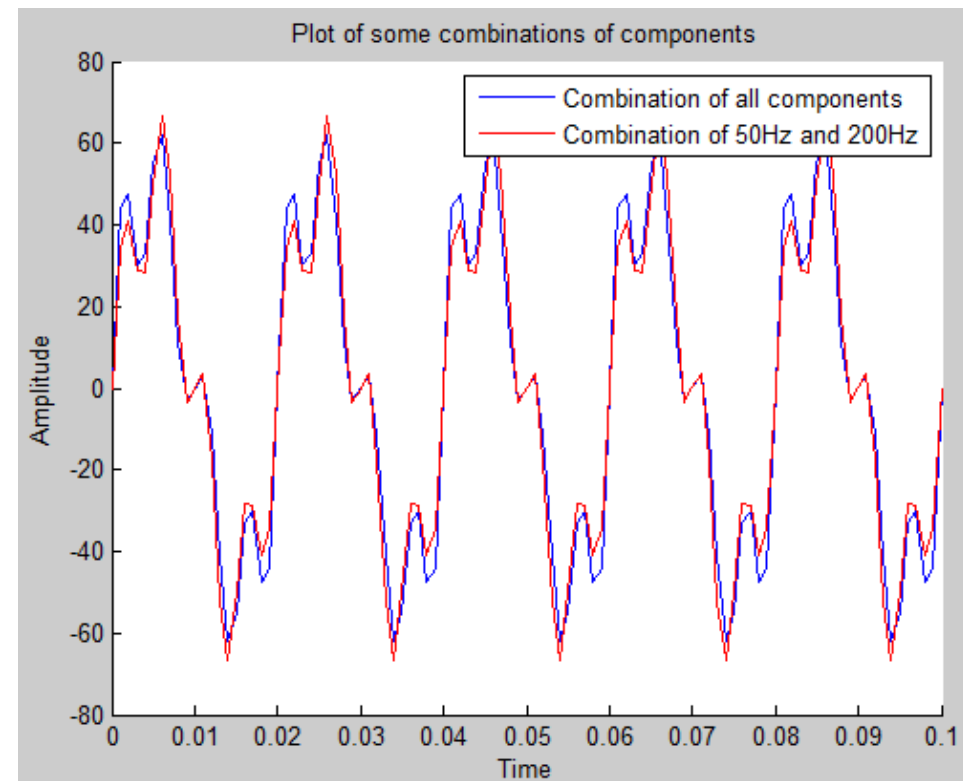
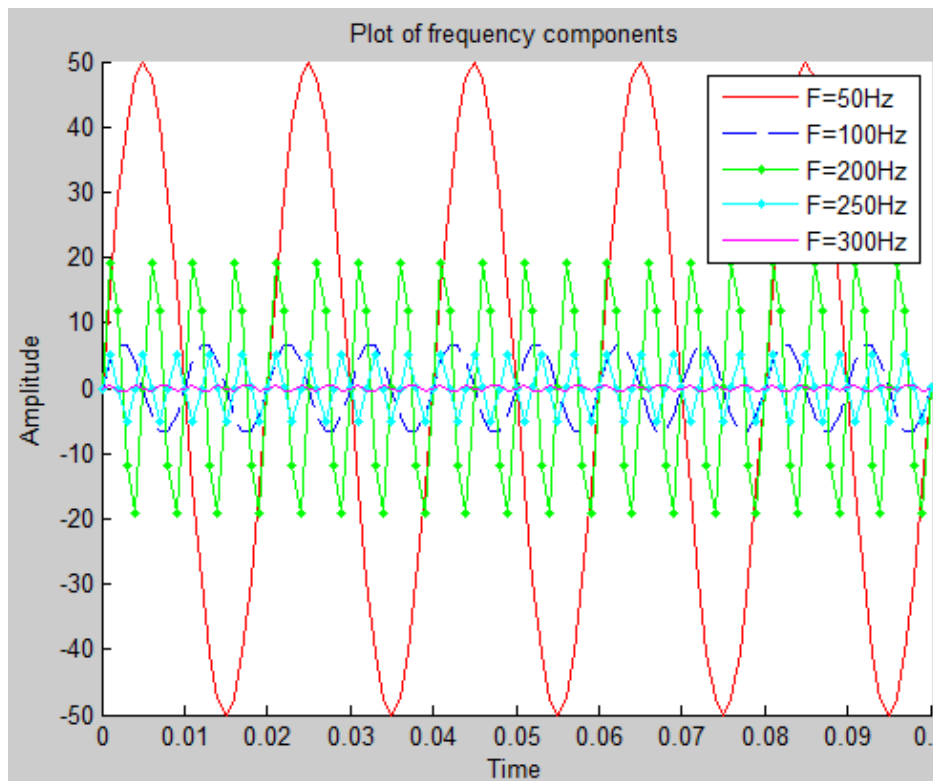
Consider the following signals

$y_k(t) = C_k * \sin(2\pi F_k t)$; where, C_k and F_k are given in the following table.

k	F_k (Hertz)	C_k
1	50	50
2	100	7
3	200	20
4	250	5
5	300	0.5

1D Discrete Fourier Transform Interpretation

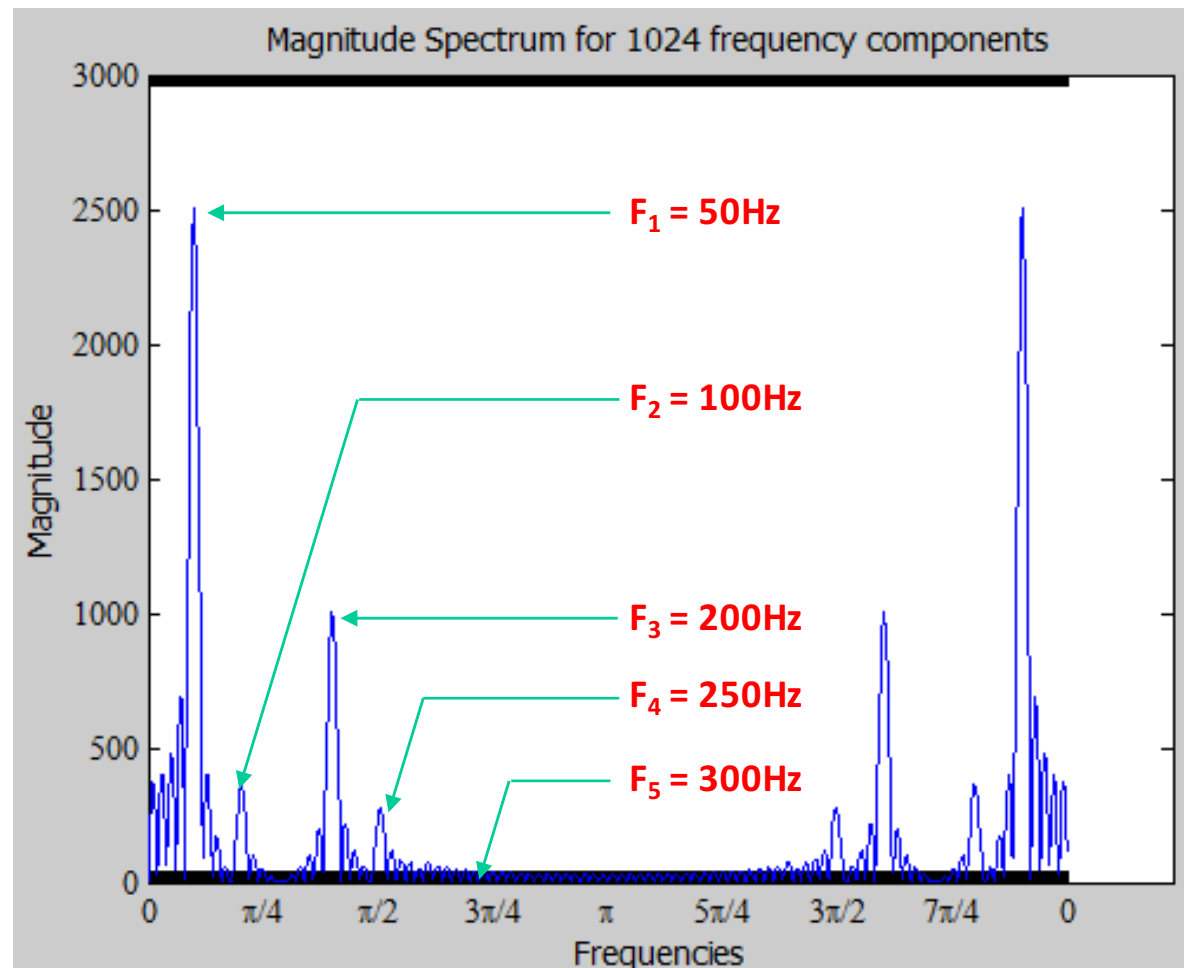
Fourier transform's meaning:



A combination of only significantly contributed components can approximate the signal that contains all the components.

1D Discrete Fourier Transform Interpretation

Fourier transform's meaning:



1D Discrete Fourier Transform

Matrix form

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j\frac{2\pi}{M}ux}$$

Example for $M = 4$

Kernel Matrix:

$$W_4 = \begin{bmatrix} e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}0} \\ e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}1} & e^{-j\frac{2\pi}{4}2} & e^{-j\frac{2\pi}{4}3} \\ e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}2} & e^{-j\frac{2\pi}{4}4} & e^{-j\frac{2\pi}{4}6} \\ e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}3} & e^{-j\frac{2\pi}{4}6} & e^{-j\frac{2\pi}{4}9} \end{bmatrix}$$

$u=0$

$u=1$

$u=2$

$u=3$



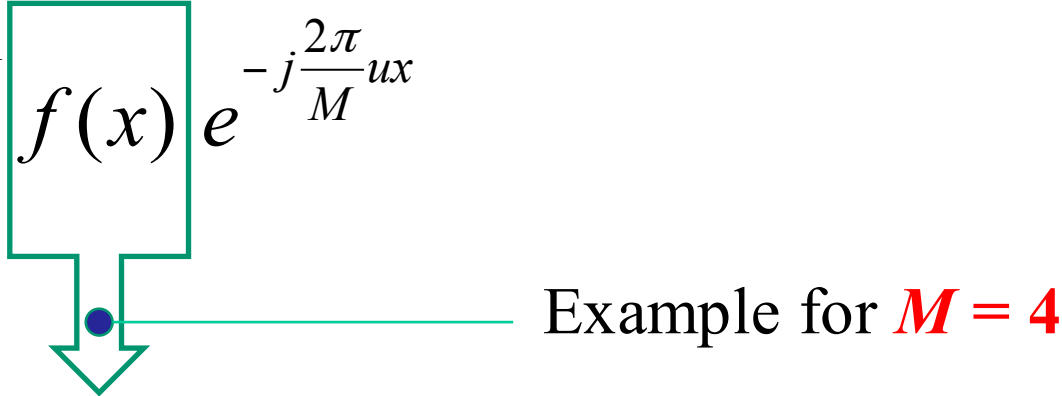
1D Discrete Fourier Transform

Matrix form

$$\begin{aligned}
 W_4 &= \begin{bmatrix} e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}0} \\ e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}1} & e^{-j\frac{2\pi}{4}2} & e^{-j\frac{2\pi}{4}3} \\ e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}2} & e^{-j\frac{2\pi}{4}4} & e^{-j\frac{2\pi}{4}6} \\ e^{-j\frac{2\pi}{4}0} & e^{-j\frac{2\pi}{4}3} & e^{-j\frac{2\pi}{4}6} & e^{-j\frac{2\pi}{4}9} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{-j\frac{\pi}{2}} & e^{-j\pi} & e^{-j\frac{3\pi}{2}} \\ 1 & e^{-j\pi} & e^{-j2\pi} & e^{-j\pi} \\ 1 & e^{-j\frac{3\pi}{2}} & e^{-j\pi} & e^{-j\frac{\pi}{2}} \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}
 \end{aligned}$$

1D Discrete Fourier Transform

Matrix form

$$F(u) = \sum_{x=0}^{M-1} \boxed{f(x)} e^{-j\frac{2\pi}{M}ux}$$


Example for $M = 4$

Vector signal:

$$f = \begin{bmatrix} 2 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

1D Discrete Fourier Transform

Matrix form

$$\begin{aligned} F(u) &= \sum_{x=0}^{M-1} f(x) e^{-j\frac{2\pi}{M}ux} \\ &= \underbrace{W_M}_{\text{Matrix}} * \underbrace{f}_{\text{Vector}} \end{aligned}$$

1D Discrete Fourier Transform

Example

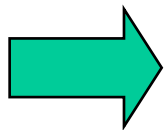
Example:

- $f(x) = \{2 \ 0 \ 3 \ 0\}$
- $M = 4$: to decompose into 4 frequency components
- $x = 0, 1, 2, 3$
- $u = 0, 1, 2, 3$
- So, $e^{-j\frac{2\pi}{M}ux}$ forms a matrix of W_4 , size of 4×4

1D Discrete Fourier Transform

Example

$$f = [2 \quad 0 \quad 3 \quad 0]^T$$



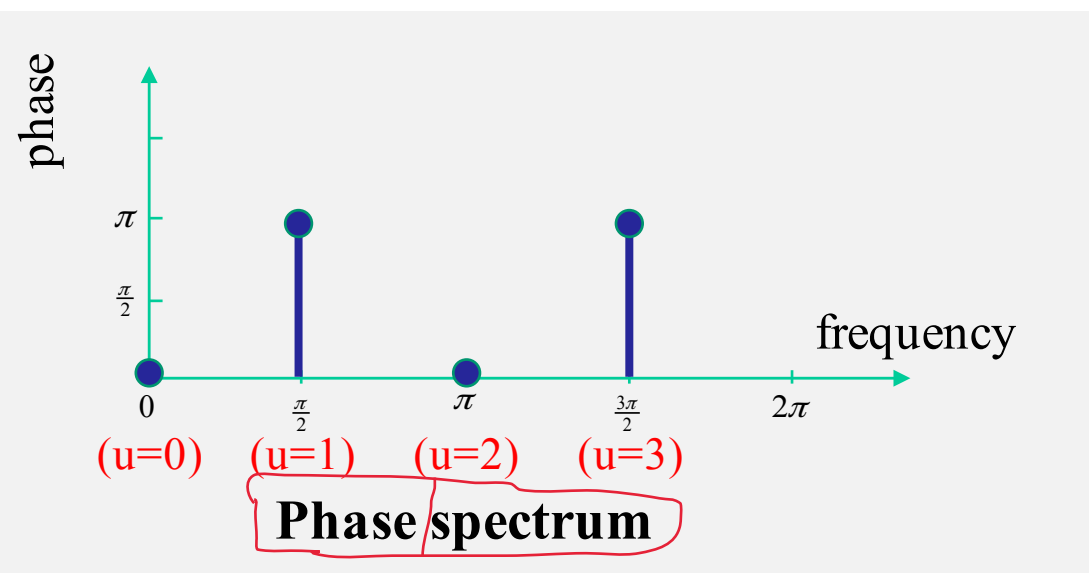
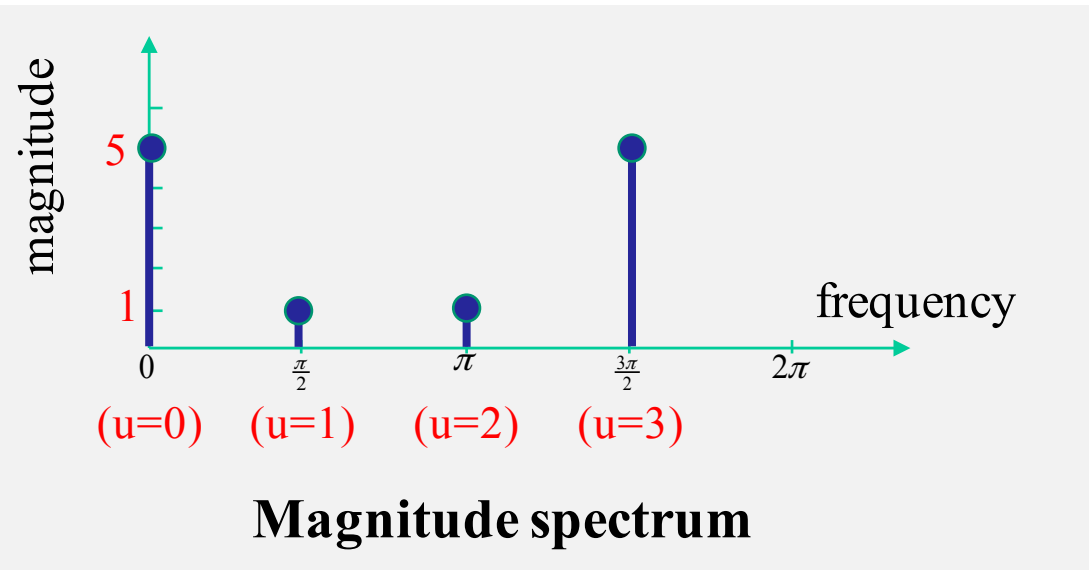
$$F(u) = W_4 f$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 5 \\ -1 \end{bmatrix}$$

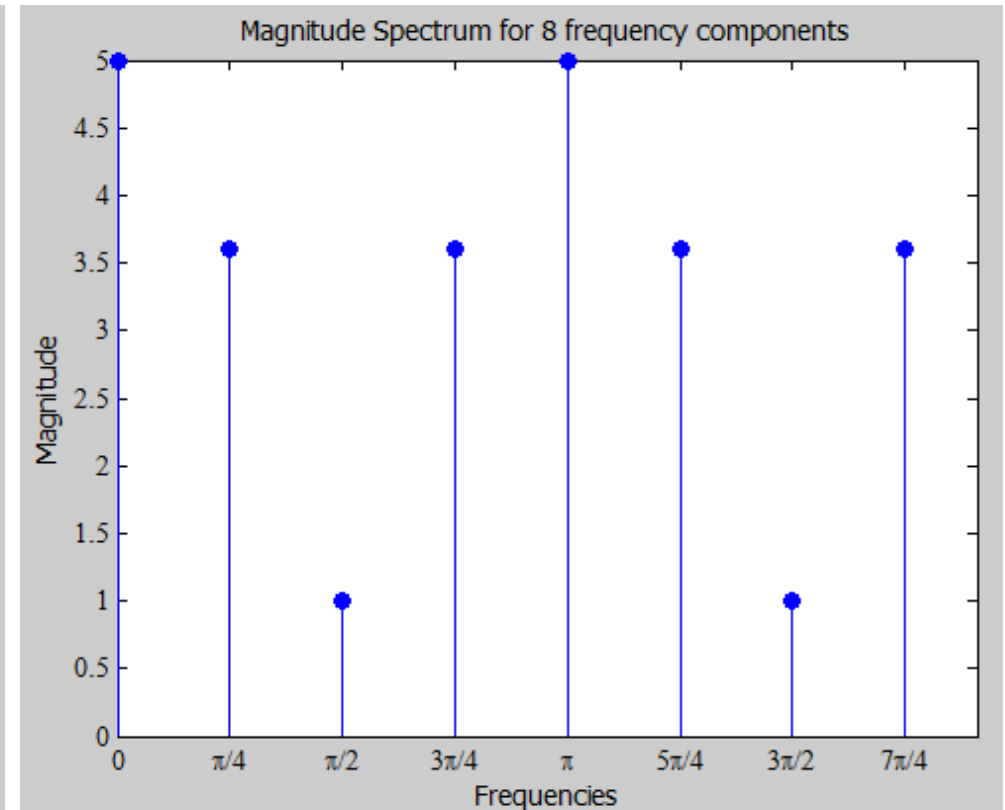
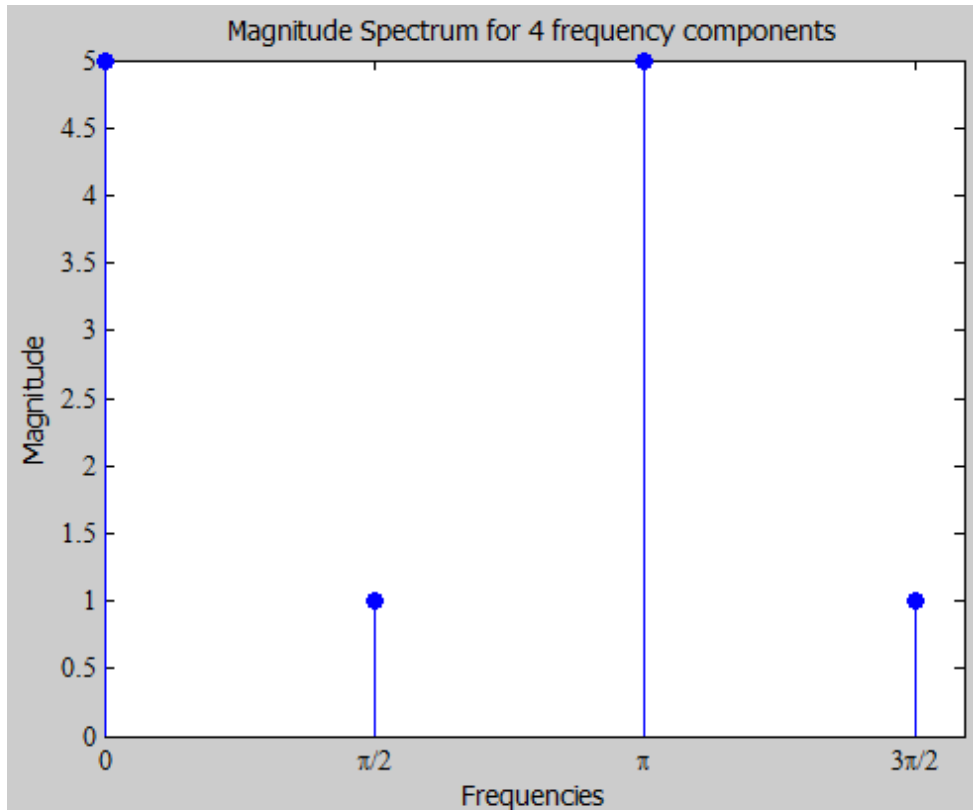
1D Discrete Fourier Transform

Example

$$F(u) = \begin{bmatrix} 5 \\ -1 \\ 5 \\ -1 \end{bmatrix}$$



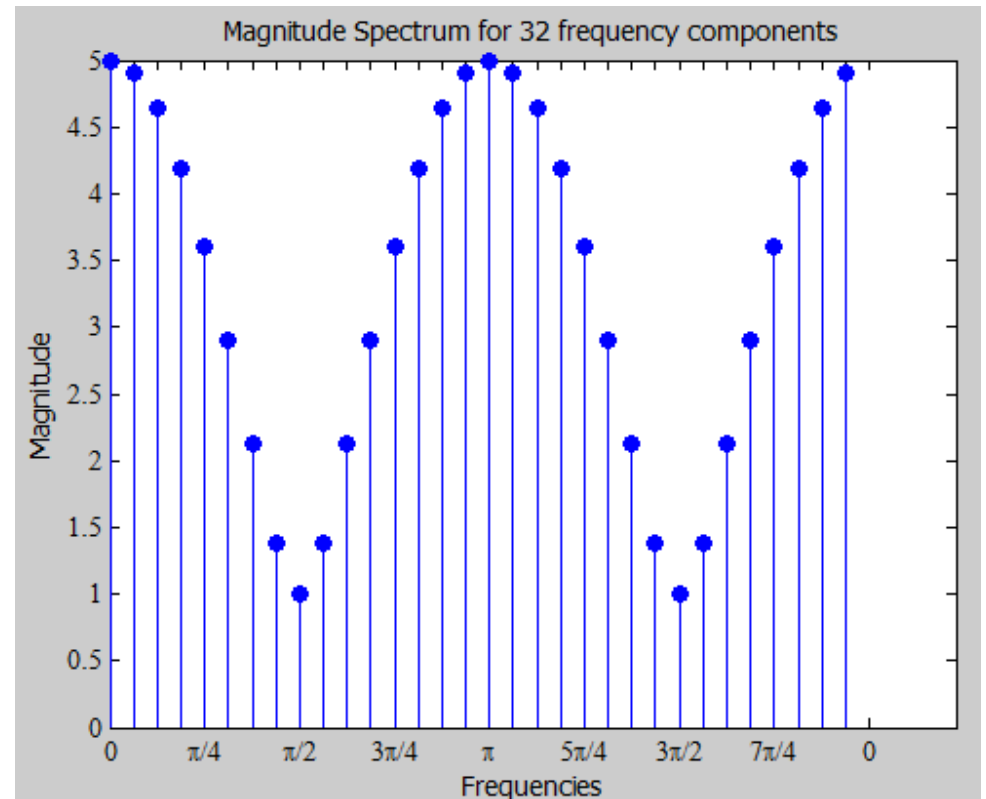
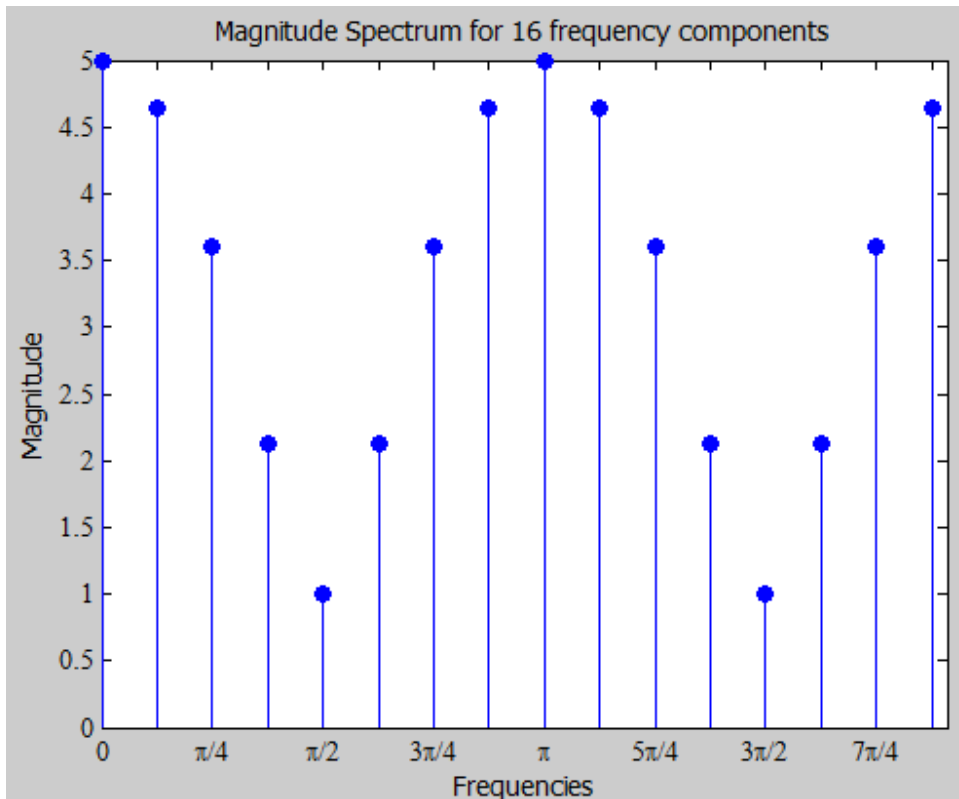
1D Discrete Fourier Transform Example



$$F(\text{🎵} = 0) = F(\text{🎵} = \text{🚗}) = 5$$

$$F(\text{🎵} = \text{🚗}/2) = F(\text{🎵} = 3 \text{ 🚗}/2) = 1$$

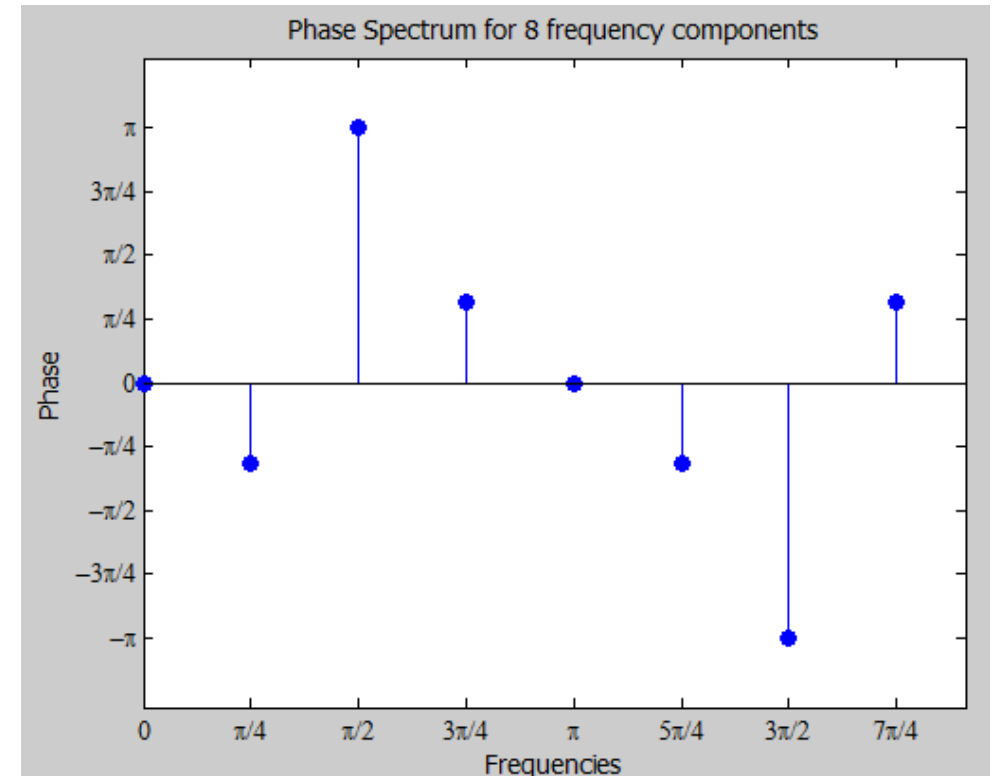
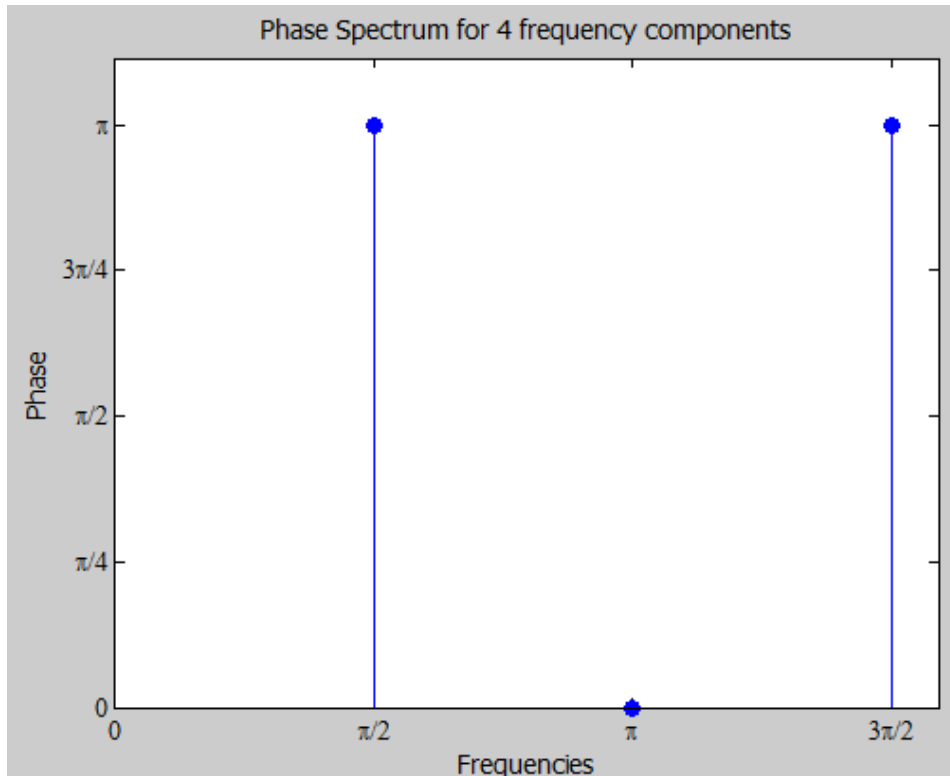
1D Discrete Fourier Transform Example



$$F(\text{🎵} = 0) = F(\text{🎵} = \text{🚗}) = 5$$

$$F(\text{🎵} = \text{🚗}/2) = F(\text{🎵} = 3 \text{ 🚗}/2) = 1$$

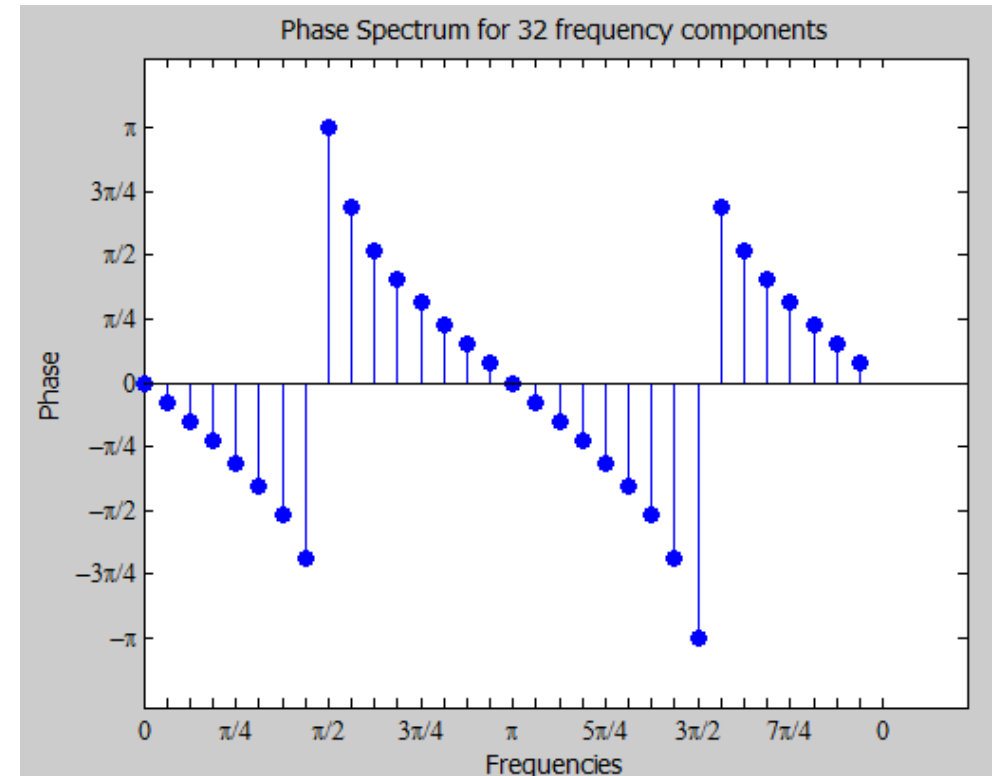
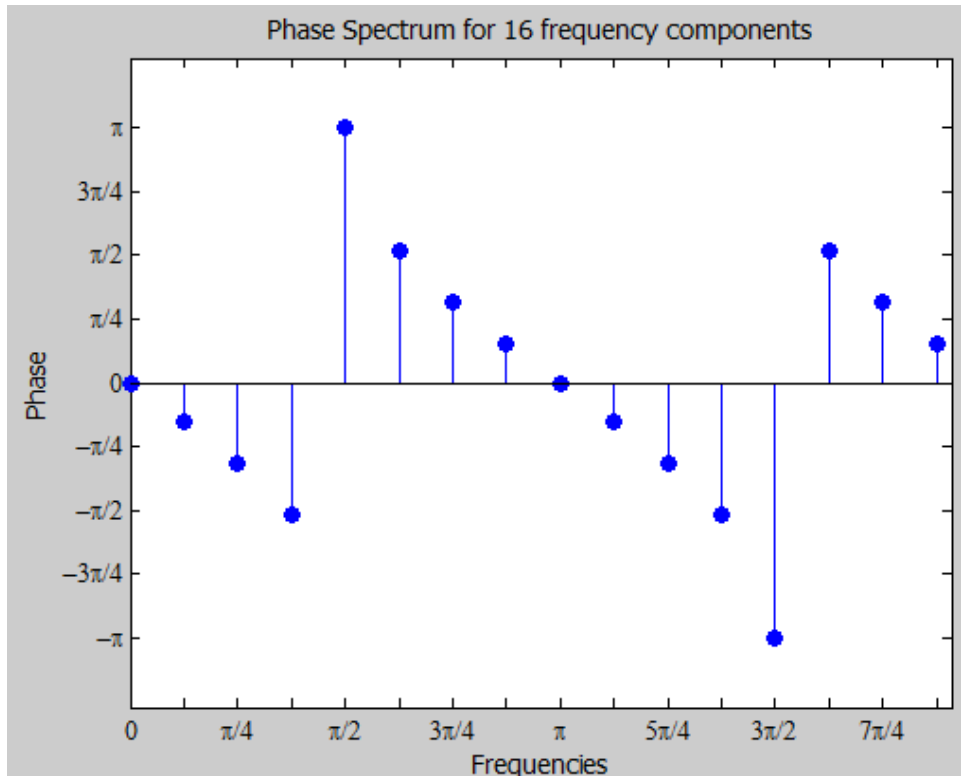
1D Discrete Fourier Transform Example



$$F(\uparrow = 0) = F(\uparrow = \text{car}) = 0 \text{ (radian)}$$

$$F(\uparrow = \text{car}/2) = F(\uparrow = 3 \text{ car}/2) = \text{car} \text{ (or - car the same value)}$$

1D Discrete Fourier Transform Example



$$F(\text{🚦} = 0) = F(\text{🚦} = \text{🚗}) = 0 \text{ (radian)}$$

$$F(\text{🚦} = \text{🚗}/2) = F(\text{🚦} = 3 \text{ 🚗}/2) = \text{🚗} \text{ (or - 🚗 the same value)}$$

1D Discrete Fourier Transform

Matrix form

Kernel Matrix:

$$W_8 = \begin{matrix} & \begin{matrix} x=0 & x=1 & x=3 & x=4 & \dots \end{matrix} \\ \begin{matrix} u=0 \\ u=1 \\ u=2 \\ u=3 \\ \vdots \end{matrix} & \begin{bmatrix} e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}0} \\ e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}1} & e^{-j\frac{\pi}{4}2} & e^{-j\frac{\pi}{4}3} & e^{-j\frac{\pi}{4}4} & e^{-j\frac{\pi}{4}5} & e^{-j\frac{\pi}{4}6} & e^{-j\frac{\pi}{4}7} \\ e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}2} & e^{-j\frac{\pi}{4}4} & e^{-j\frac{\pi}{4}6} & e^{-j\frac{\pi}{4}8} & e^{-j\frac{\pi}{4}10} & e^{-j\frac{\pi}{4}12} & e^{-j\frac{\pi}{4}14} \\ e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}3} & e^{-j\frac{\pi}{4}6} & e^{-j\frac{\pi}{4}9} & e^{-j\frac{\pi}{4}12} & e^{-j\frac{\pi}{4}15} & e^{-j\frac{\pi}{4}18} & e^{-j\frac{\pi}{4}21} \\ e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}4} & e^{-j\frac{\pi}{4}8} & e^{-j\frac{\pi}{4}12} & e^{-j\frac{\pi}{4}16} & e^{-j\frac{\pi}{4}20} & e^{-j\frac{\pi}{4}24} & e^{-j\frac{\pi}{4}28} \\ e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}5} & e^{-j\frac{\pi}{4}10} & e^{-j\frac{\pi}{4}15} & e^{-j\frac{\pi}{4}20} & e^{-j\frac{\pi}{4}25} & e^{-j\frac{\pi}{4}30} & e^{-j\frac{\pi}{4}35} \\ e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}6} & e^{-j\frac{\pi}{4}12} & e^{-j\frac{\pi}{4}18} & e^{-j\frac{\pi}{4}24} & e^{-j\frac{\pi}{4}30} & e^{-j\frac{\pi}{4}36} & e^{-j\frac{\pi}{4}42} \\ e^{-j\frac{\pi}{4}0} & e^{-j\frac{\pi}{4}7} & e^{-j\frac{\pi}{4}14} & e^{-j\frac{\pi}{4}21} & e^{-j\frac{\pi}{4}28} & e^{-j\frac{\pi}{4}35} & e^{-j\frac{\pi}{4}42} & e^{-j\frac{\pi}{4}49} \end{bmatrix} \end{matrix}$$

1D Discrete Fourier Transform

Matrix form

Kernel Matrix:

$$W_8 = \begin{matrix} & \begin{matrix} x=0 & x=1 & x=3 & x=4 & \dots \end{matrix} & \\ \begin{matrix} u=0 \\ u=1 \\ u=2 \\ u=3 \\ \vdots \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & e^{-j\frac{\pi}{4}} & e^{-j\frac{\pi}{2}} & e^{-j\frac{3\pi}{4}} & e^{-j\pi} & e^{-j\frac{5\pi}{4}} & e^{-j\frac{3\pi}{2}} & e^{-j\frac{7\pi}{4}} \\ 1 & e^{-j\frac{\pi}{2}} & e^{-j\pi} & e^{-j\frac{3\pi}{2}} & e^{-j2\pi} & e^{-j\frac{\pi}{2}} & e^{-j\pi} & e^{-j\frac{3\pi}{2}} \\ 1 & e^{-j\frac{3\pi}{4}} & e^{-j\frac{3\pi}{2}} & e^{-j\frac{\pi}{4}} & e^{-j\pi} & e^{-j\frac{7\pi}{4}} & e^{-j\frac{\pi}{2}} & e^{-j\frac{5\pi}{4}} \\ 1 & e^{-j\pi} & e^{-j2\pi} & e^{-j\pi} & e^{-j2\pi} & e^{-j\pi} & e^{-j2\pi} & e^{-j\pi} \\ 1 & e^{-j\frac{5\pi}{4}} & e^{-j\frac{\pi}{2}} & e^{-j\frac{7\pi}{4}} & e^{-j\pi} & e^{-j\frac{\pi}{4}} & e^{-j\frac{3\pi}{2}} & e^{-j\frac{3\pi}{4}} \\ 1 & e^{-j\frac{3\pi}{2}} & e^{-j\pi} & e^{-j\frac{\pi}{2}} & e^{-j2\pi} & e^{-j\frac{3\pi}{2}} & e^{-j\pi} & e^{-j\frac{\pi}{2}} \\ 1 & e^{-j\frac{7\pi}{4}} & e^{-j\frac{3\pi}{2}} & e^{-j\frac{5\pi}{4}} & e^{-j\pi} & e^{-j\frac{3\pi}{4}} & e^{-j\frac{\pi}{2}} & e^{-j\frac{\pi}{4}} \end{bmatrix} \end{matrix}$$

1D Discrete Fourier Transform

Matrix form

Kernel Matrix:

$$W_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{\sqrt{2}}{2}(1-j) & -j & -\frac{\sqrt{2}}{2}(1+j) & -1 & -\frac{\sqrt{2}}{2}(1-j) & j & \frac{\sqrt{2}}{2}(1+j) \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -\frac{\sqrt{2}}{2}(1+j) & j & \frac{\sqrt{2}}{2}(1-j) & -1 & \frac{\sqrt{2}}{2}(1+j) & -j & -\frac{\sqrt{2}}{2}(1-j) \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{\sqrt{2}}{2}(1-j) & -j & \frac{\sqrt{2}}{2}(1+j) & -1 & \frac{\sqrt{2}}{2}(1-j) & j & -\frac{\sqrt{2}}{2}(1+j) \\ 1 & j & -1 & -j & 1 & -j & -1 & -j \\ 1 & \frac{\sqrt{2}}{2}(1+j) & j & -\frac{\sqrt{2}}{2}(1+j) & -1 & -\frac{\sqrt{2}}{2}(1+j) & -j & \frac{\sqrt{2}}{2}(1-j) \end{bmatrix}$$

1D Discrete Fourier Transform

Matrix form

Kernel Matrix's Properties:

□ Symmetric on the main diagonal

□ Let $c_M^u = e^{-j\frac{2\pi}{M}u}$
Then

$$c_M^{u+\frac{M}{2}} = -c_M^u$$

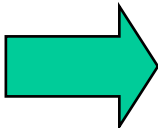
$$c_M^{u+M} = c_M^u$$

$$\begin{aligned} f &= \frac{u}{M} \\ e^{-i \cdot 2\pi f} \\ &= \cos(2\pi f) \end{aligned}$$

1D Discrete Fourier Transform

Question

Signal: $f = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}$

 $F(u) = W_8 f$

$= ?$

1. Draw magnitude and phase spectrum for $F(u)$
2. How much does frequency $u = 3$ contribute to the signal (in magnitude and in phase)?
3. Write Matlab/C/CPP function program to compute DFT/FFT.
4. Let $M_1 = 256$ and $M_2 = 8$ be number of frequency into which we decompose, and $F_{256}(u)$ and $F_8(u)$ are Fourier coefficients for those two transforms. Assume that we already have $F_{256}(u)$, how can obtain $F_8(u)$ from $F_{256}(u)$?

2D Discrete Fourier Transform

Formula

Forward transform:

$$F(u, v) = \sum_{y=0}^{N-1} \sum_{x=0}^{M-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$u=0..M-1 \quad v=0..N-1$

Backward transform:

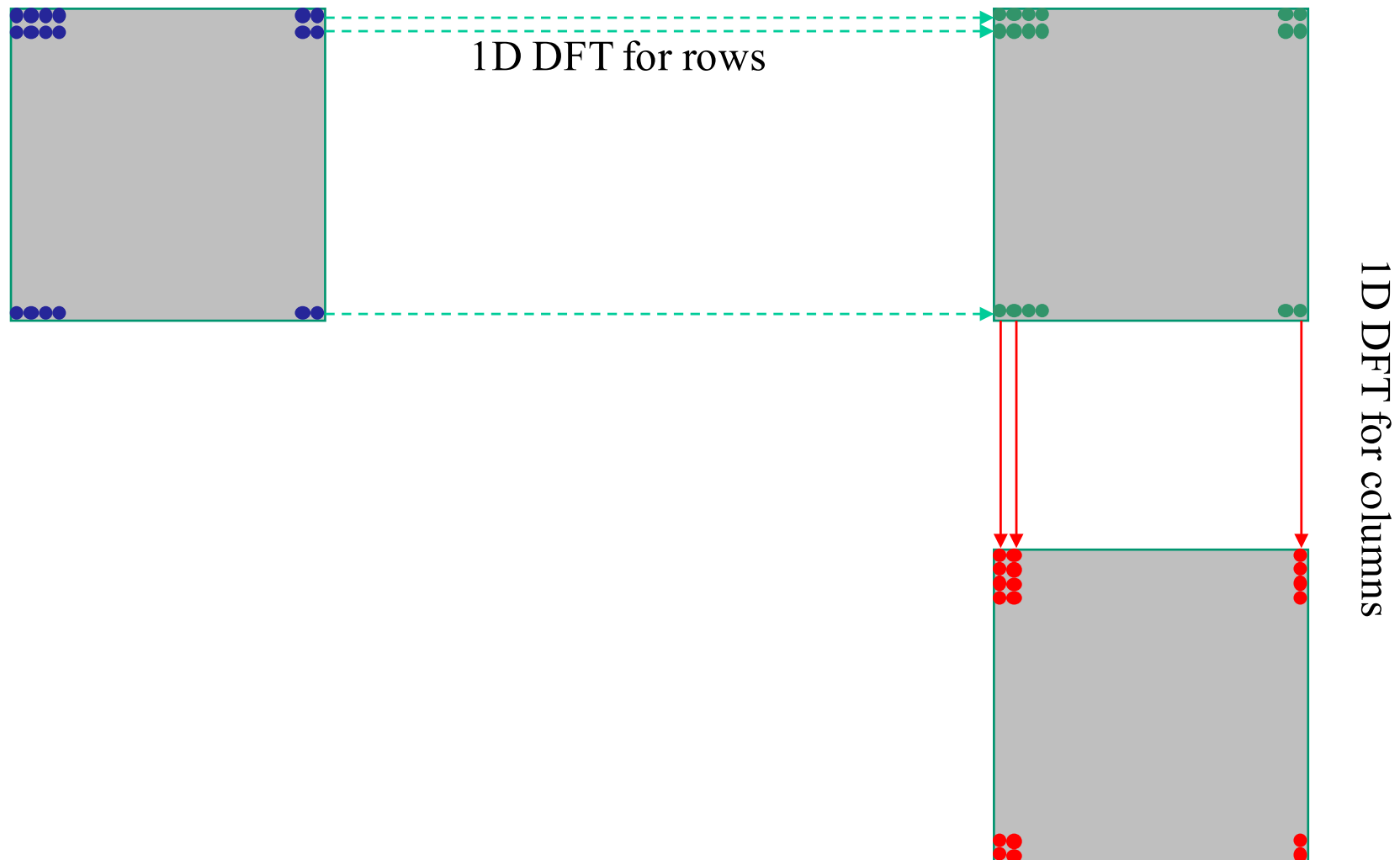
$$f(x, y) = \sum_{v=0}^{N-1} \sum_{u=0}^{M-1} F(u, v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$x=0..M-1 \quad y=0..N-1$

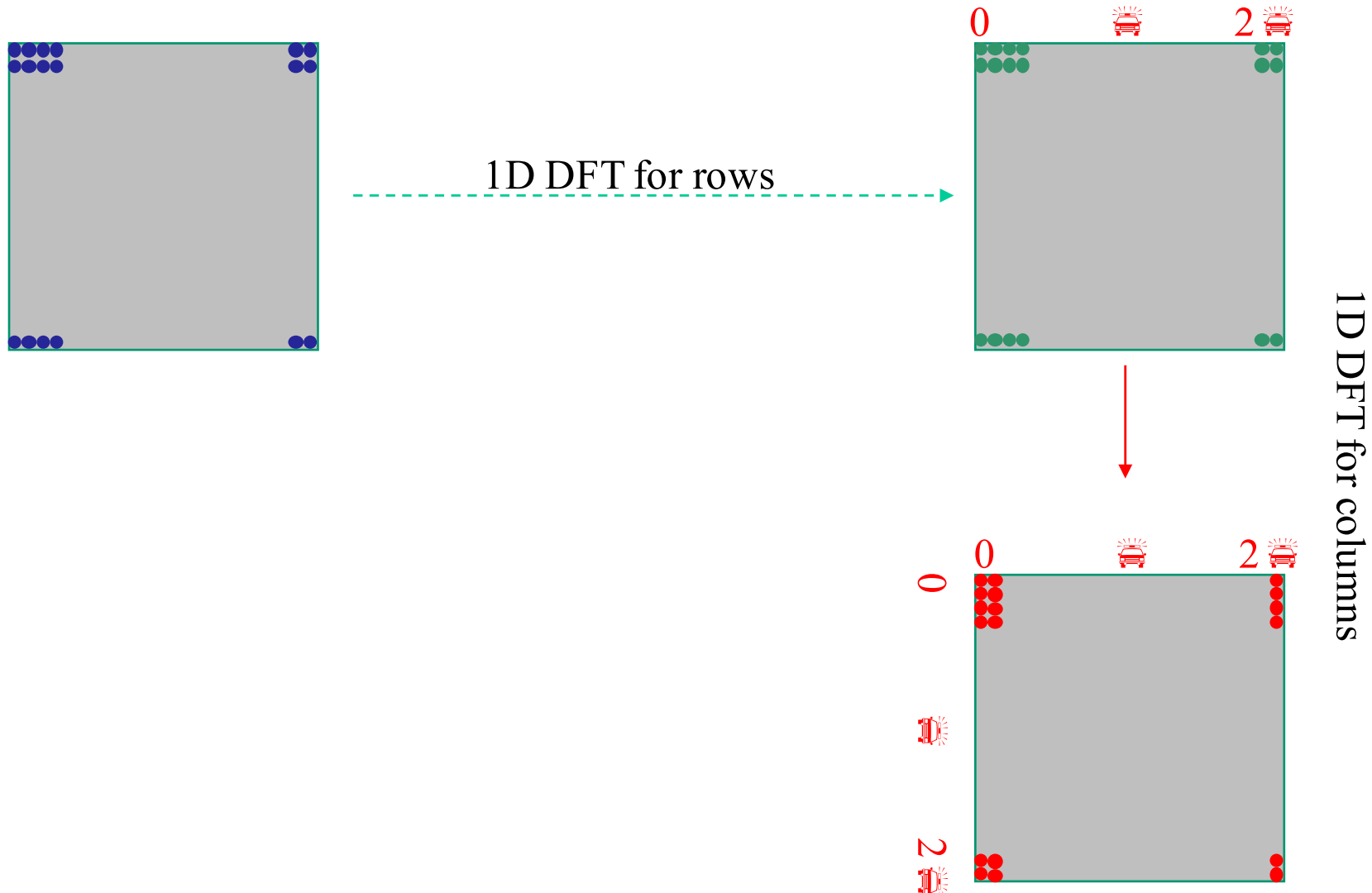
2D Discrete Fourier Transform Computation

$$\begin{aligned} F(u, v) &= \sum_{y=0}^{N-1} \sum_{x=0}^{M-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \\ &= \sum_{y=0}^{N-1} \underbrace{\sum_{x=0}^{M-1} f(x, y) e^{-j\frac{2\pi}{M}ux} e^{-j\frac{2\pi}{N}vx}}_{\text{1D DFT for rows}} \\ &= \underbrace{\sum_{y=0}^{N-1} F(u, y) e^{-j\frac{2\pi}{N}vx}}_{\text{1D DFT for columns}} \end{aligned}$$

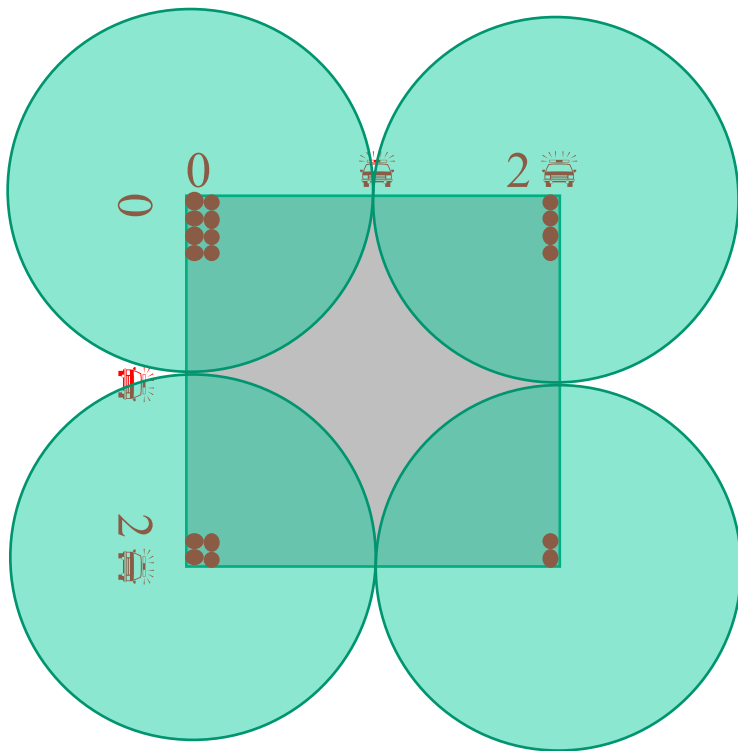
2D Discrete Fourier Transform Computation



2D Discrete Fourier Transform Computation

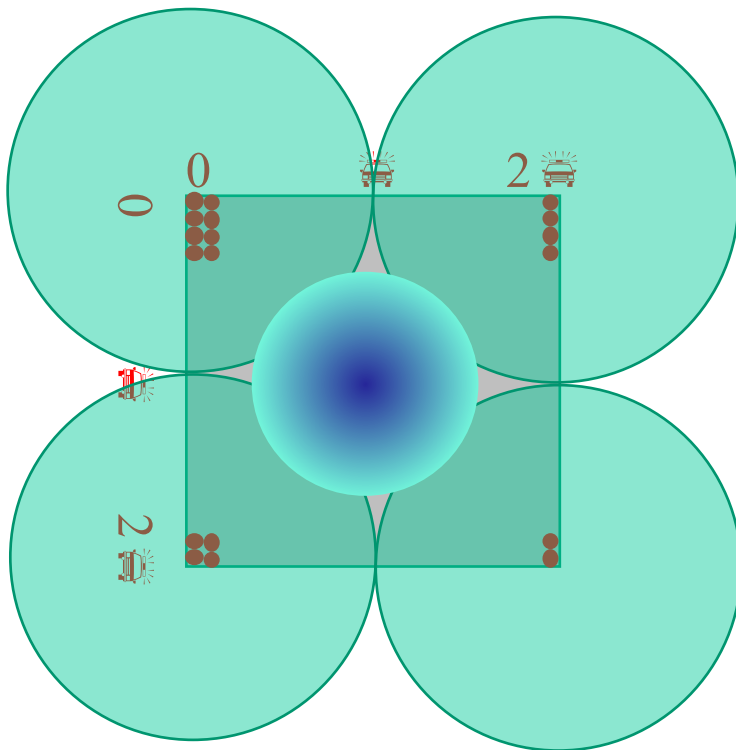


2D Discrete Fourier Transform Computation



- ❖ Low frequencies are at four corners
- ❖ High frequencies are at the center
- ❖ Low frequencies mean intensities inside the input images have small variation. For example, perfectly smooth image has only one frequency, 0 radian.

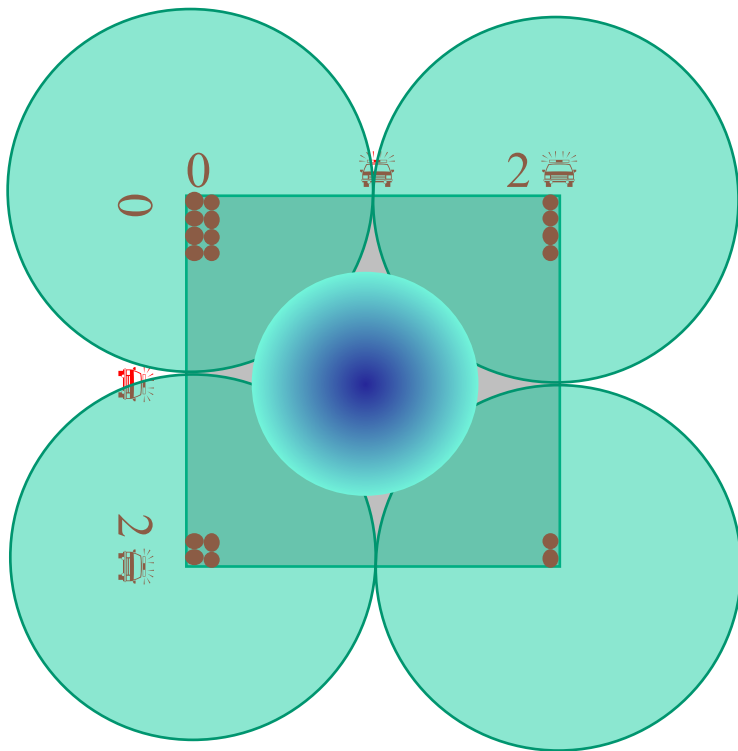
2D Discrete Fourier Transform Computation



shift

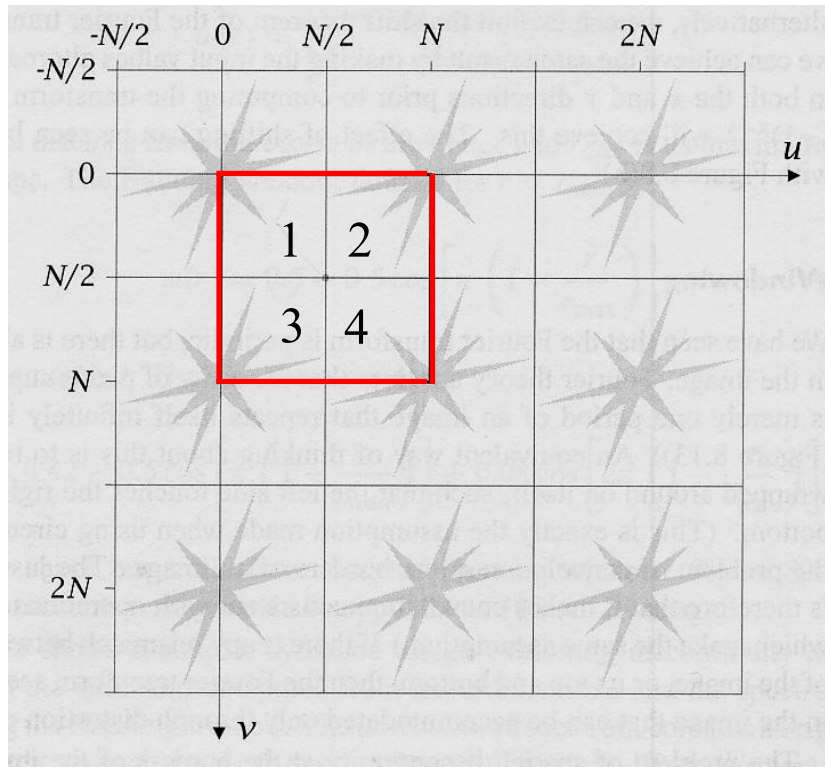
- ❖ Low frequencies are at four corners
- ❖ High frequencies are at the center
- ❖ High frequencies mean the image has a large variation. For example, images have high variation around their edges.

2D Discrete Fourier Transform Computation



- ❖ DFT are periodic on both of horizontal and vertical directions, as shown in the next slide
- ❖ So, SHIFT (or swap) the diagonal direction to place low frequencies at the center of spectrum.
- ❖ Why shift? Coefficients (or contribution) of low frequencies are usually significant larger than high frequencies'

Fourier Transform property (1)



(*: conjugate)

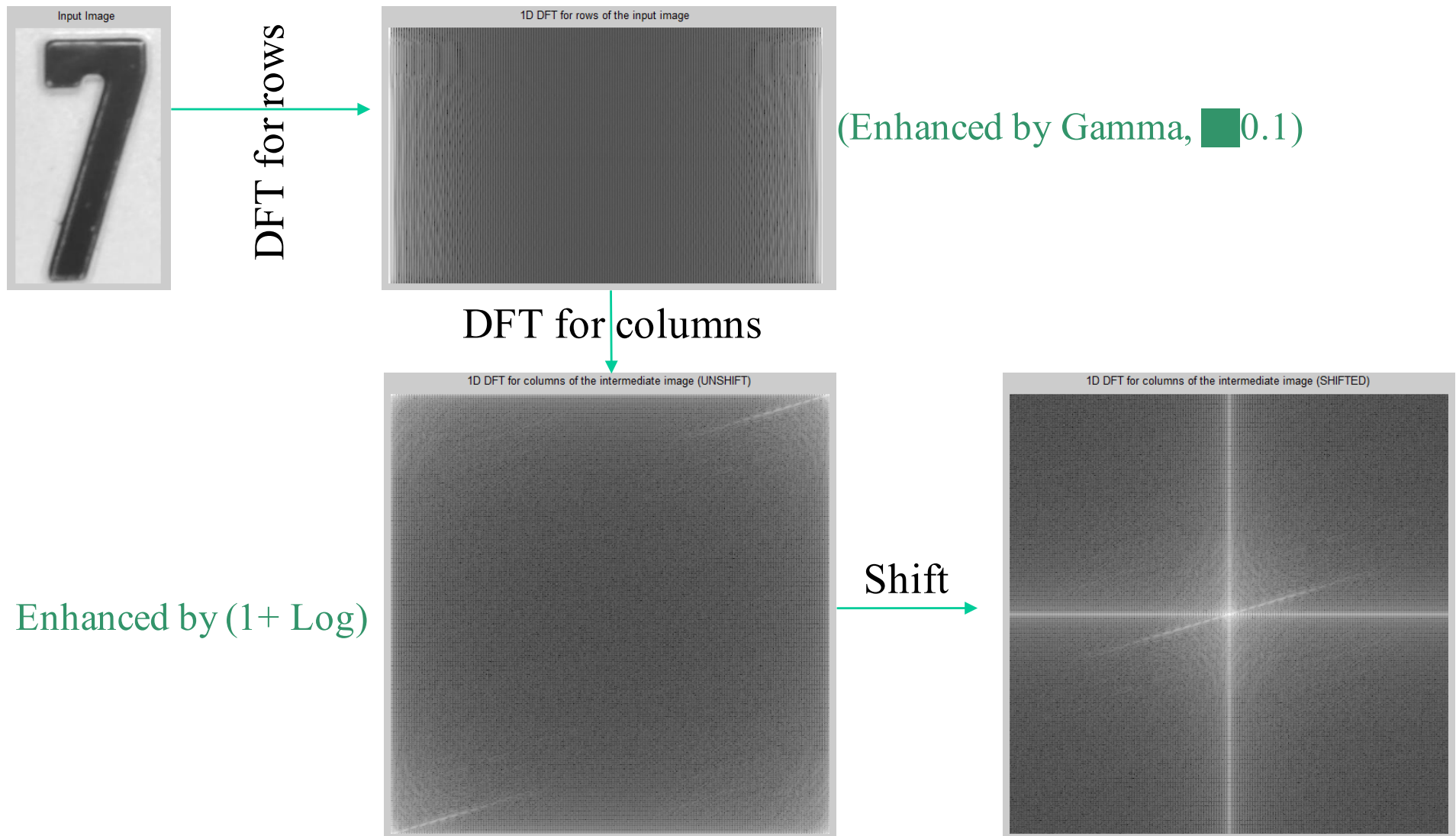
$$1 = 4^*$$

$$2 = 3^*$$

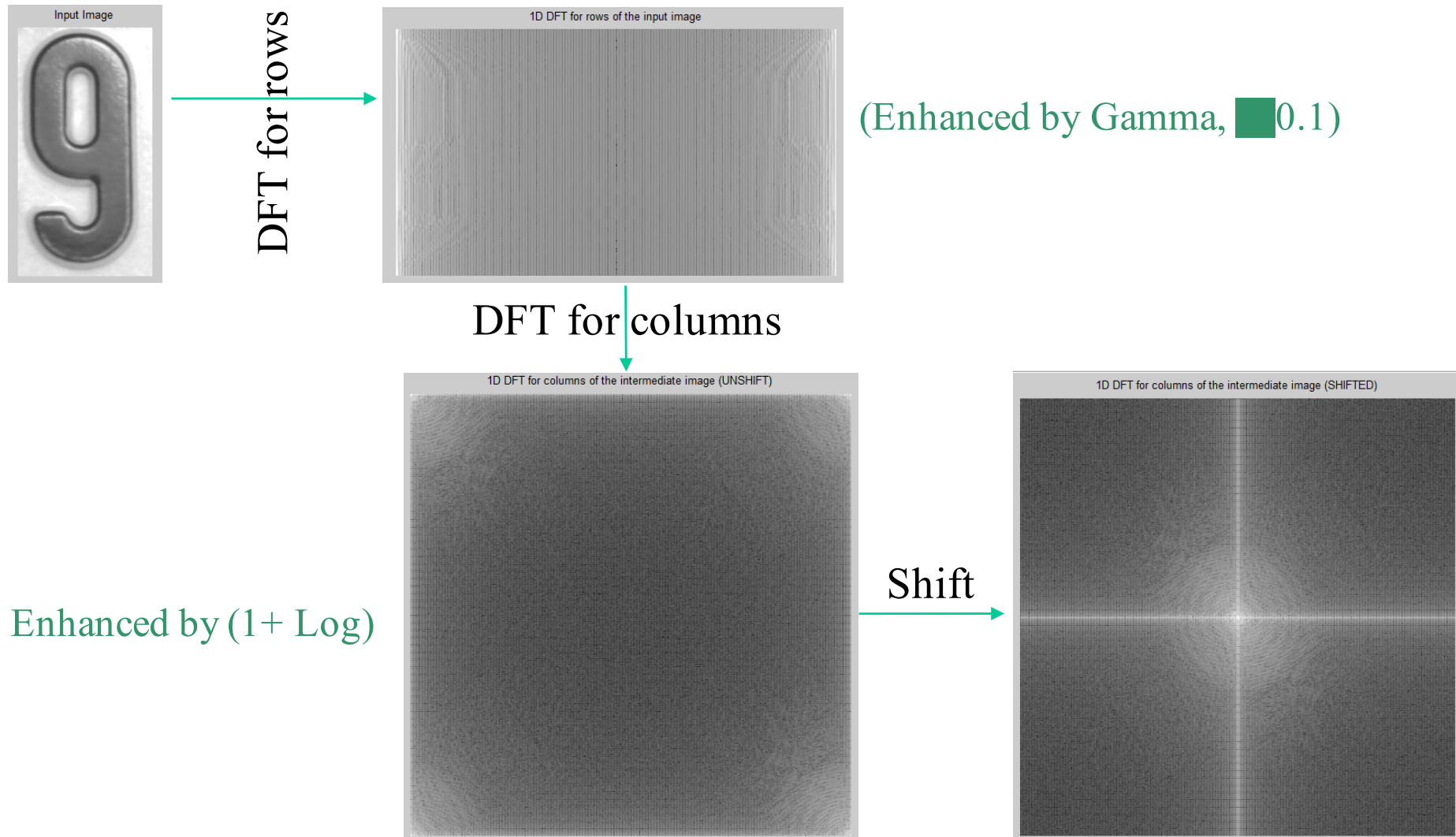
$$F(u, v) = F(-u, -v)^*$$

$$F(u, v) = F(u + M, v) = F(u, v + N) = F(u + M, v + N)$$

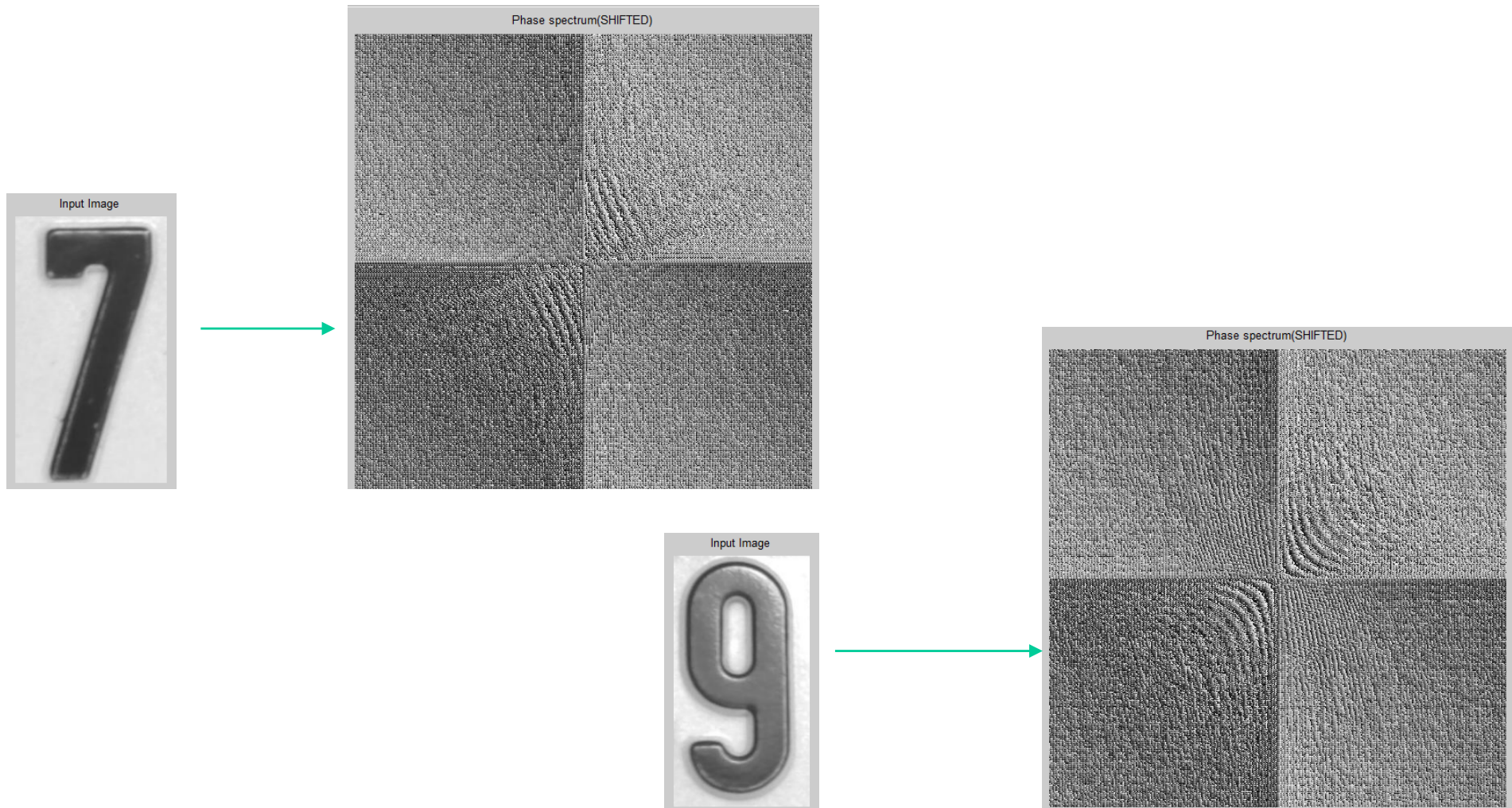
2D Discrete Fourier Transform Computation



2D Discrete Fourier Transform Computation



2D Discrete Fourier Transform Computation



Fast Fourier Transform Computation

$$\begin{aligned}
 W_M^{u \underline{2x}} &= e^{-j \frac{2\pi}{2N} 2ux} \\
 &= e^{-j \frac{2\pi}{N} ux} \\
 &= W_N^{ux}
 \end{aligned}$$

$$\begin{aligned}
 W_M^{u \underline{(2x+1)}} &= e^{-j \frac{2\pi}{2N} u(2x+1)} \\
 &= e^{-j \frac{\pi}{N} (2ux+u)} \\
 &= e^{-j \frac{2\pi}{N} ux} e^{-j \frac{2\pi}{2N} u} \\
 &= e^{-j \frac{2\pi}{M} u} e^{-j \frac{2\pi}{N} ux} \\
 &= W_M^{u \uparrow} W_N^{ux}
 \end{aligned}$$

$$x(2x) = x(x)$$

$$x(2x+1) = x(1) \cdot x(x)$$

Fast Fourier Transform Computation

$$\begin{aligned}
 F(u) &= \sum_{x=0}^{M-1} f(x) w_M^{ux} \\
 &= \sum_{x=0}^{N-1} f(2x) w_M^{u2x} + \sum_{x=0}^{N-1} f(2x+1) w_M^{u(2x+1)} \\
 &= \underbrace{\sum_{x=0}^{N-1} f(2x) w_N^{ux}}_{F_{\text{even}}(u)} + w_M^u \underbrace{\sum_{x=0}^{N-1} f(2x+1) w_N^{ux}}_{F_{\text{odd}}(u)}
 \end{aligned}$$

$\downarrow$ $u = 0..M-1$
 $ = 0..2N-1$

$\downarrow$ $u = 0..N-1$

$\downarrow$ $u = 0..N-1$

Fast Fourier Transform Computation

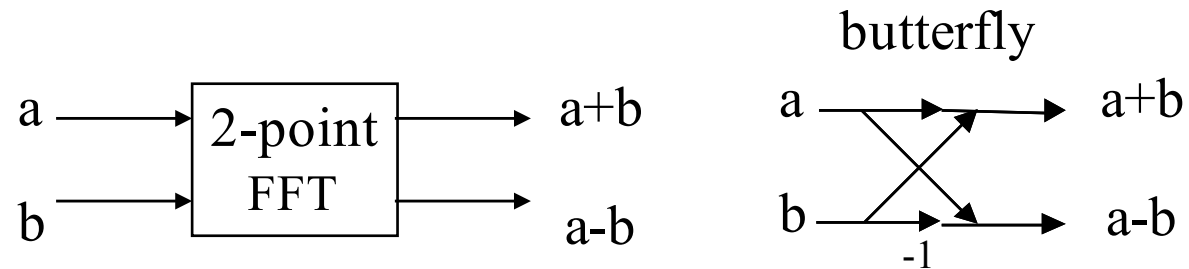
$$\begin{aligned}w_M^{u+N} &= e^{-j\frac{2\pi}{2N}(u+N)} \\&= e^{-j\frac{\pi}{N}(u+N)} \\&= e^{-j\pi} e^{-j\frac{2\pi}{2N}u} \\&= -w_M^u\end{aligned}$$

Fast Fourier Transform Computation

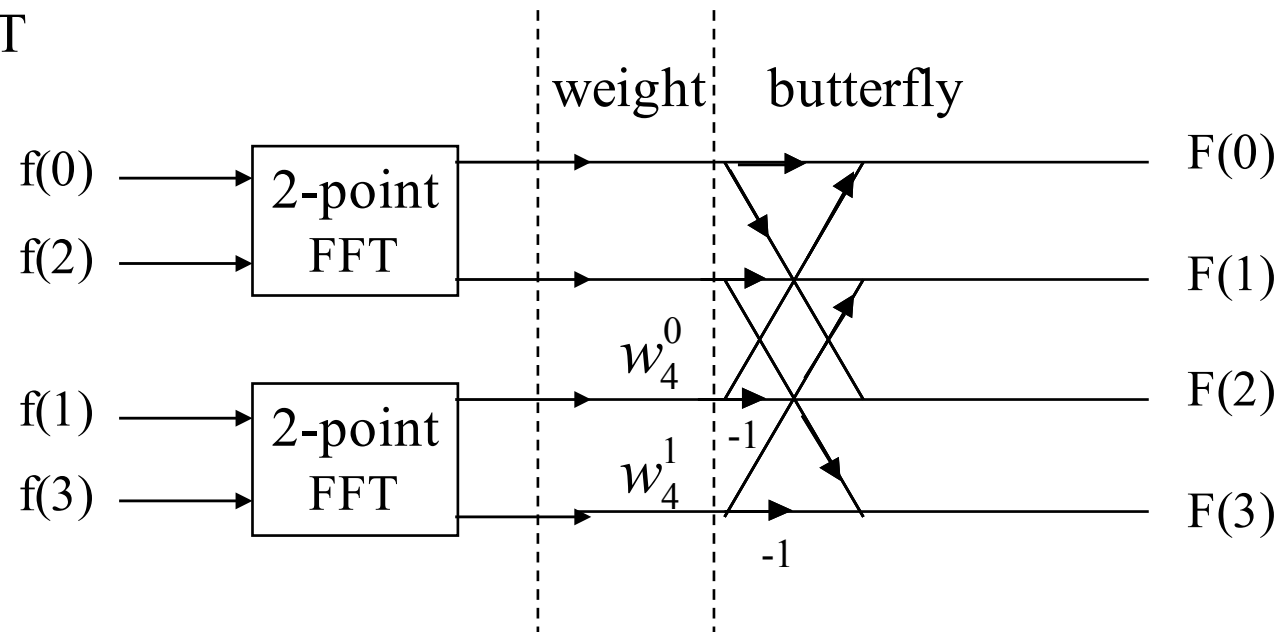
$$F(u) = \begin{cases} F_{even}(u) + w_M^u F_{odd}(u) & | u = 0..(N-1) \\ F_{even}(u) - w_M^u F_{odd}(u) & | u = N..(2M-1) \end{cases}$$

DIT-FFT (Scalable) (1)

2-point FFT

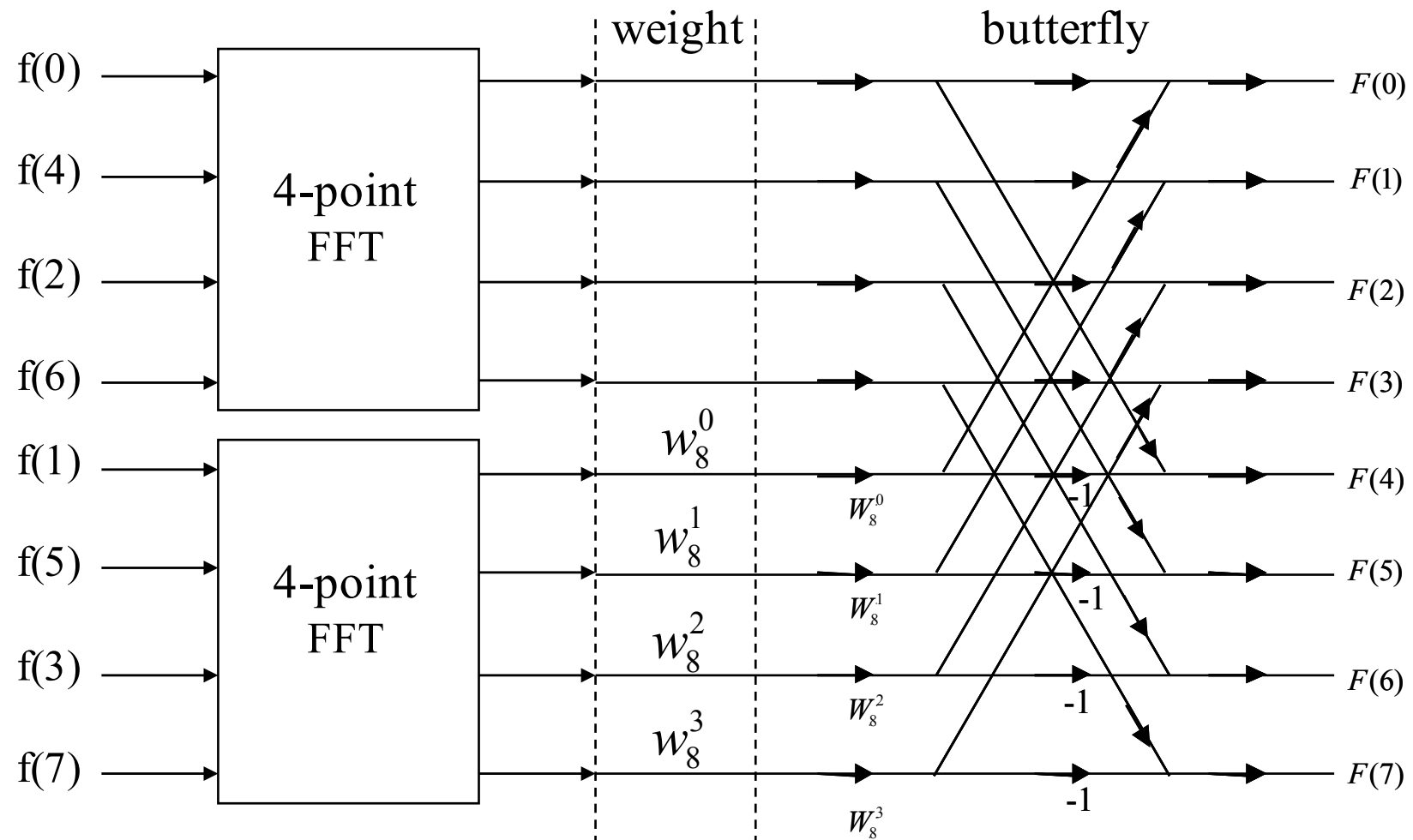


4-point FFT



DIT-FFT (Scalable) (2)

8-point FFT

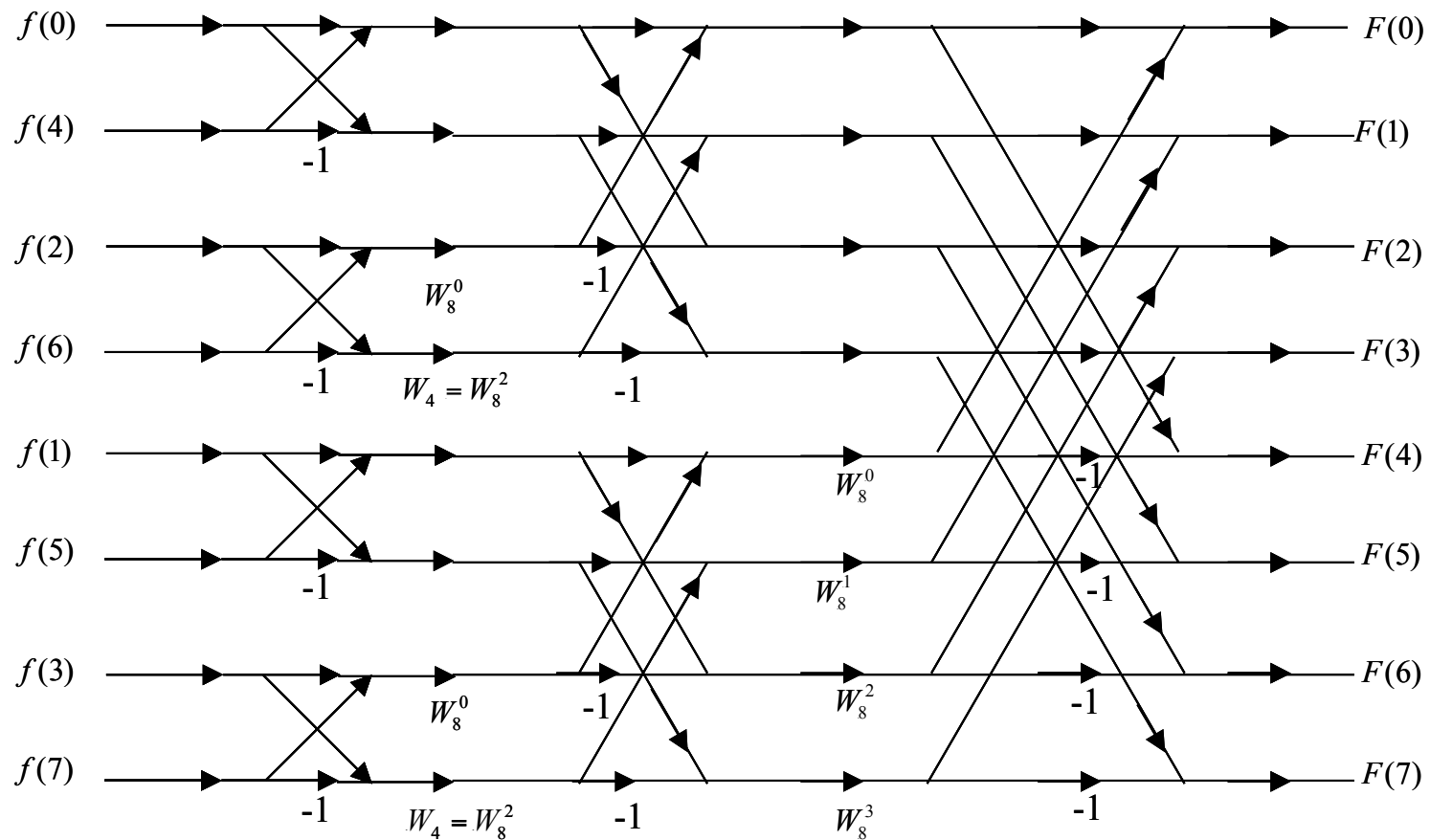


Fast Fourier Transform (DIT-FFT)

Direct calculation = N^2

FFT = $N \log_2 N$

8-point FFT



Fast Fourier Transform

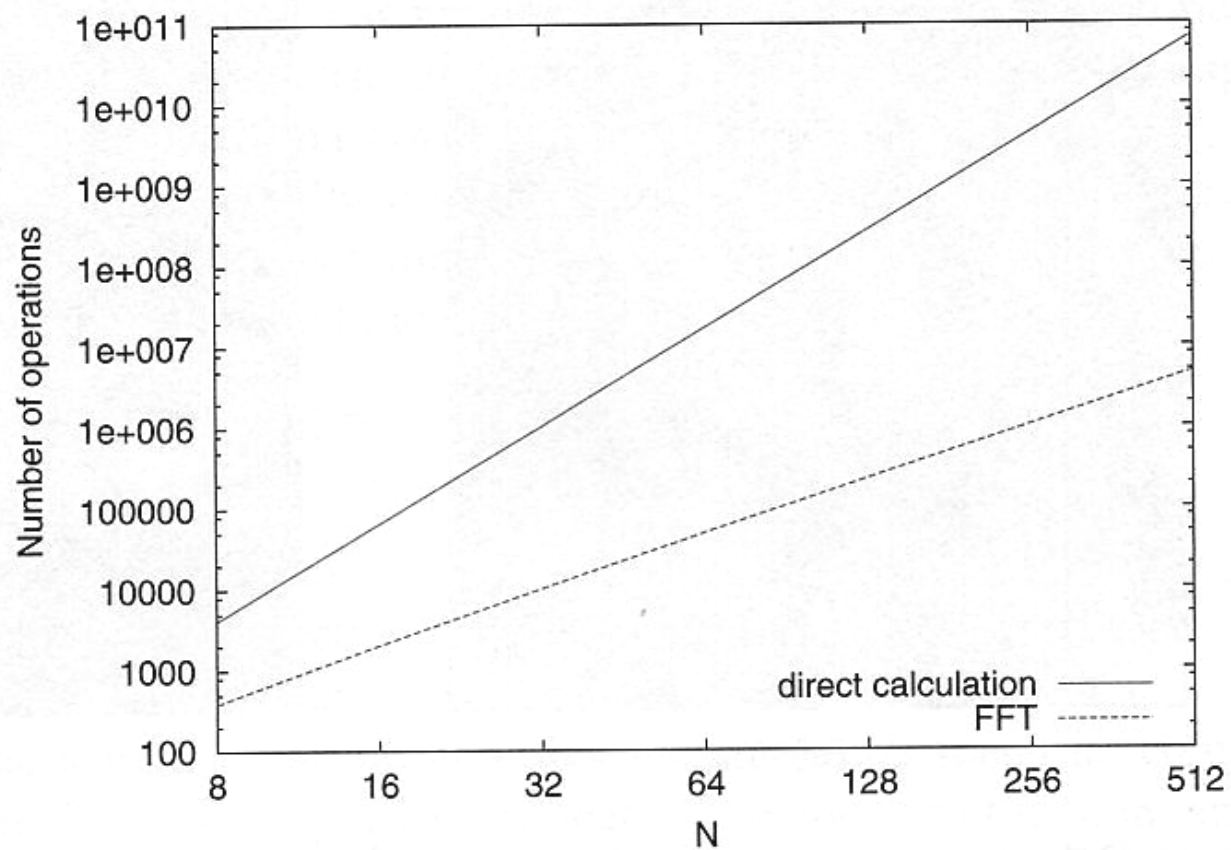
Computational Complexity:

Discrete Fourier Transform $\rightarrow O(N^2)$

Fast Fourier Transform $\rightarrow O(N \log N)$

Remember: The Fast Fourier Transform is just a faster **algorithm** for computing the Discrete Fourier Transform — it does *not* produce a different result.

Processing Time (DFT vs FFT)



Relationship between Convolution and DFT

$$f(x, y) * h(x, y) \Leftrightarrow F(u, v)H(u, v)$$

$$g(x, y) = f(x, y) * h(x, y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n)$$

$$\begin{aligned} DFT[g(x, y)] = G(u, v) &= \sum_{y=0}^{M-1} \sum_{x=0}^{N-1} g(x, y) W_M^{ux} W_N^{vy} ; & W_M^{ux} &= e^{-j\frac{2\pi ux}{M}} ; W_N^{vy} = e^{-j\frac{2\pi vy}{N}} \\ &= \sum_{y=0}^{M-1} \sum_{x=0}^{N-1} \left[\frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n) \right] W_M^{ux} W_N^{vy} \\ &= \frac{1}{MN} \sum_{y=0}^{M-1} \sum_{x=0}^{N-1} \left[\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x - m, y - n) \right] W_M^{mu} W_N^{nv} W_M^{(x-m)u} W_N^{(y-n)v} \\ &= \frac{1}{MN} \left[\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) W_M^{mu} W_N^{nv} \right] \sum_{y=0}^{M-1} \sum_{x=0}^{N-1} h(x - m, y - n) W_M^{(x-m)u} W_N^{(y-n)v} \\ &= \frac{1}{MN} F(u, v)H(u, v) \end{aligned}$$

$$IDFT \left[\frac{1}{MN} F(u, v)H(u, v) \right] = g(x, y) = f(x, y) * h(x, y)$$

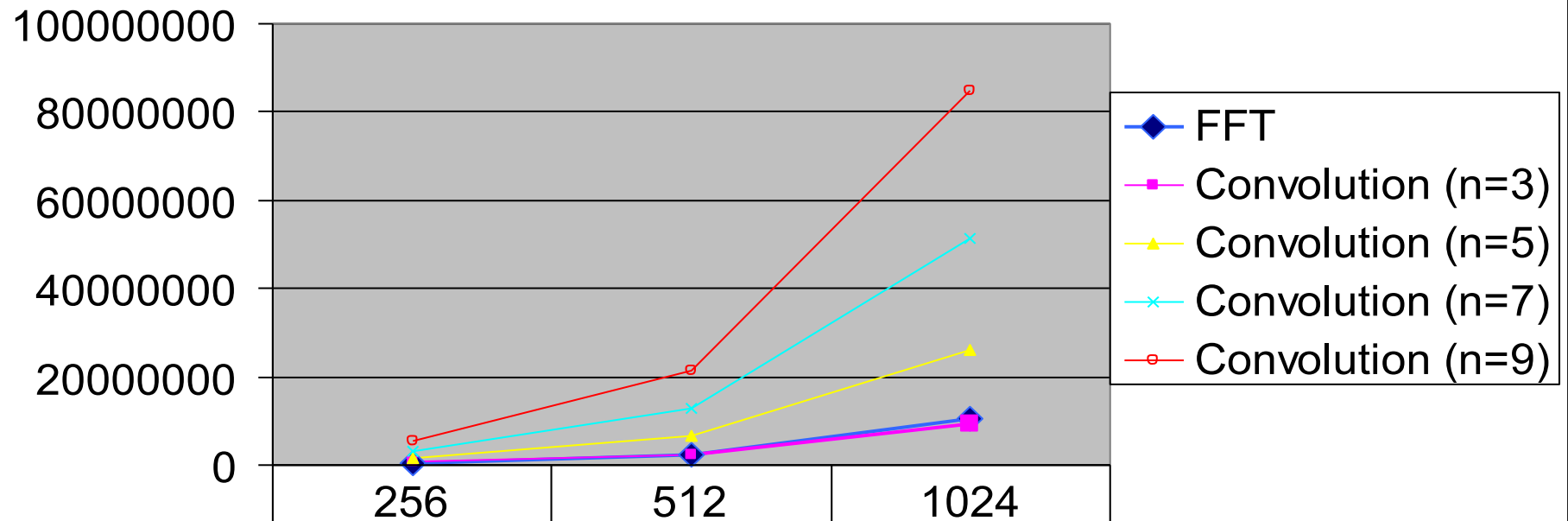
Comparison of Convolution and FFT

	Convolution	FFT
Computational Process	Simple	Complex with many steps
Number of Multiplications	$N^2 n^2$	$< N^2 (\log_2 N)$

$N \times N \rightarrow$ image size

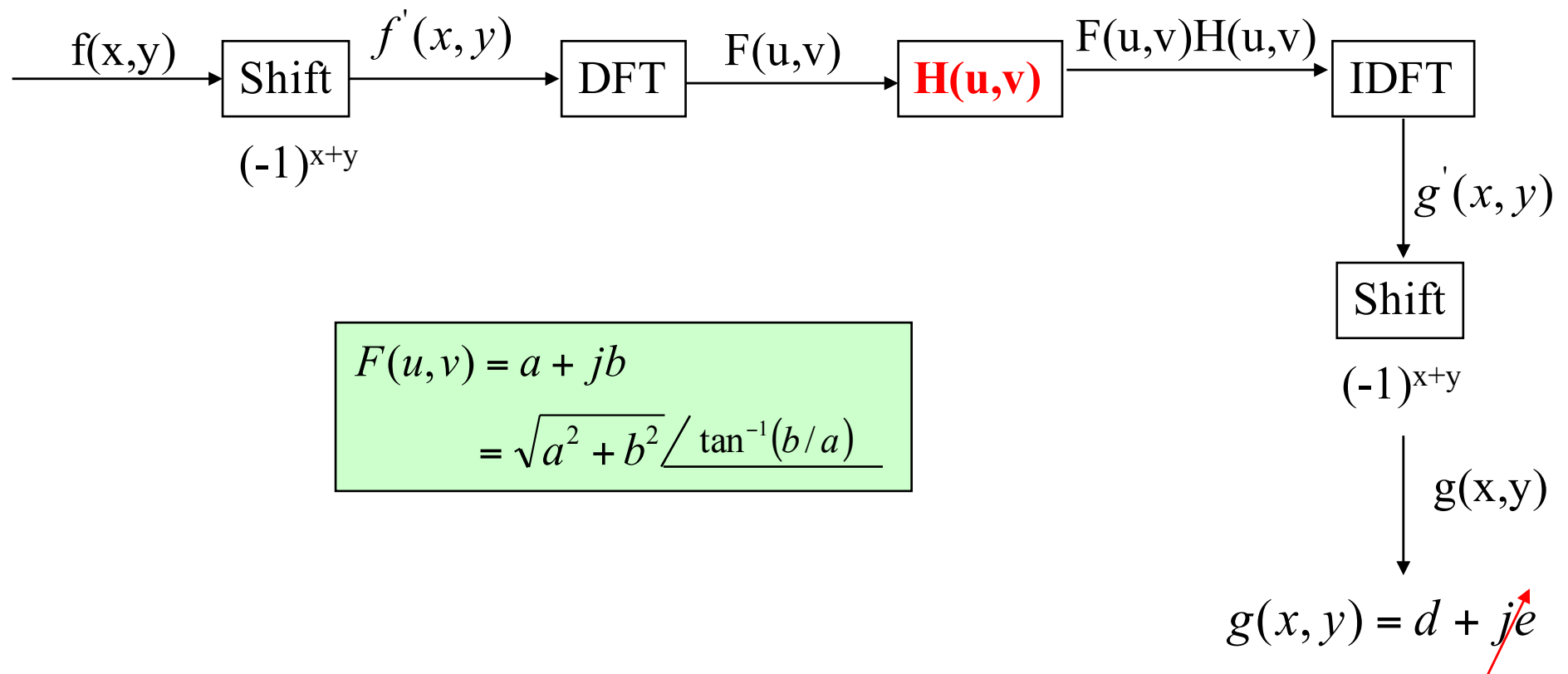
$n \times n \rightarrow$ filter window size

Number of Multiplication (FFT vs Convolution)

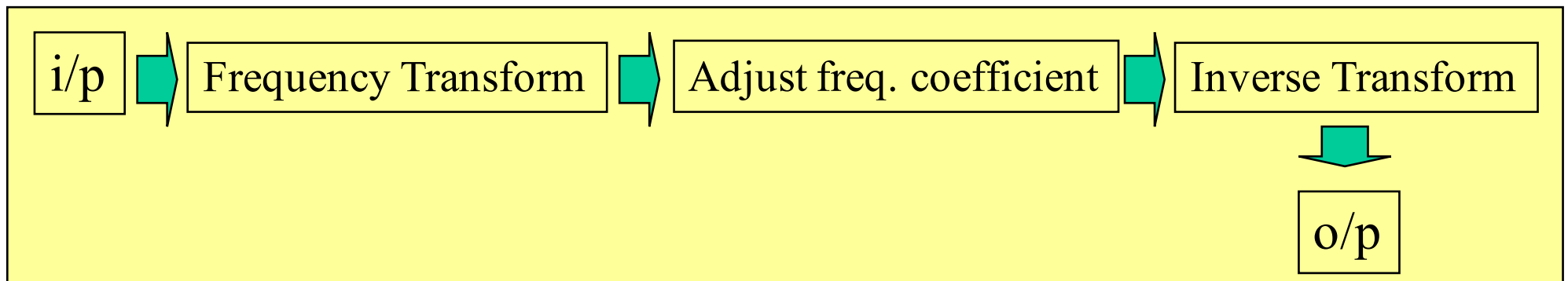
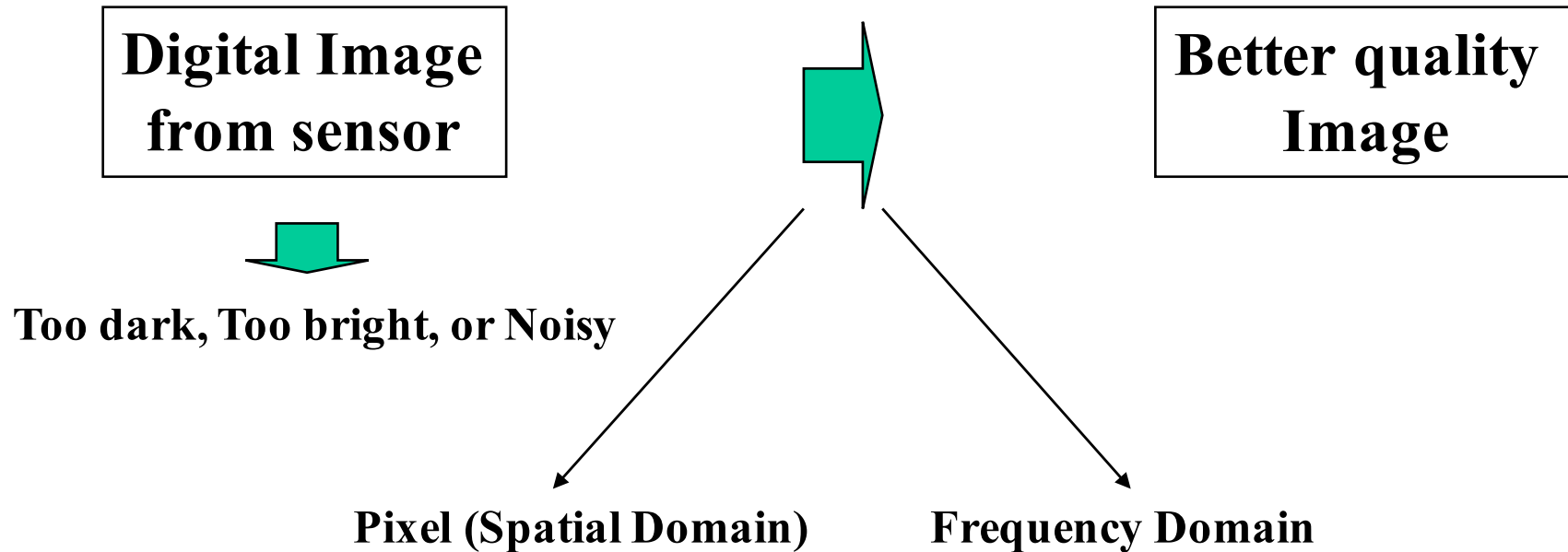


◆ FFT	524288	2359296	10485760
■ Convolution (n=3)	589824	2359296	9437184
▲ Convolution (n=5)	1638400	6553600	26214400
✕ Convolution	3211264	12845056	51380224

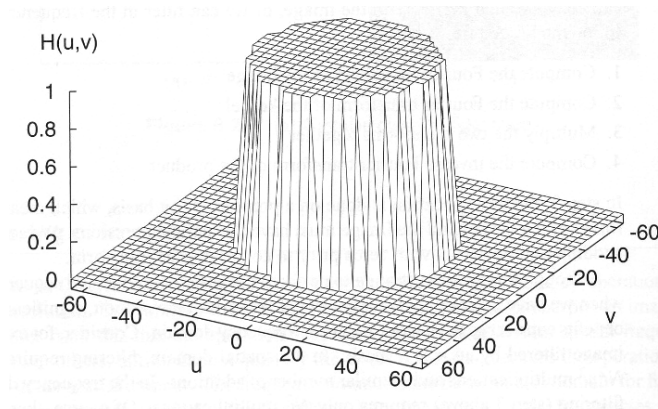
Filtering in Frequency Domain



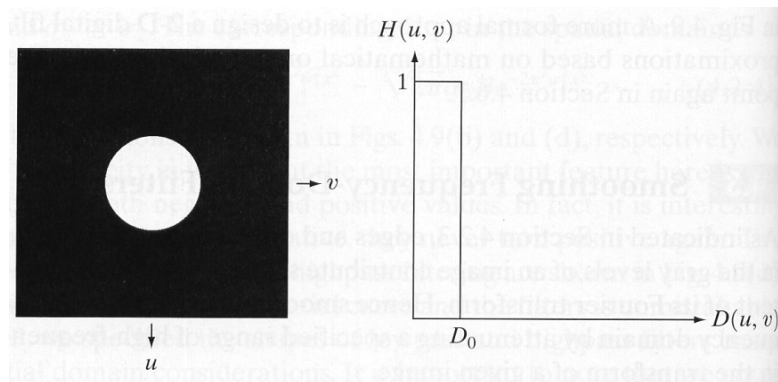
Fourier Transform Applications: Image Enhancement and Restoration



Ideal Low Pass Filter (ILPF)

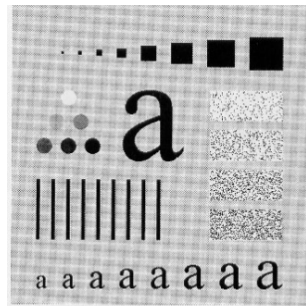


$$H(u, v) = \begin{cases} 1 & r(u, v) \leq r_0 \\ 0 & r(u, v) > r_0 \end{cases}$$

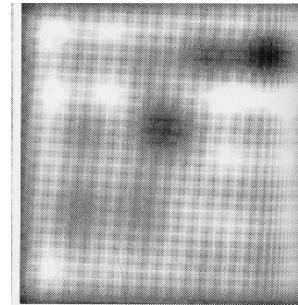


$$r(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$

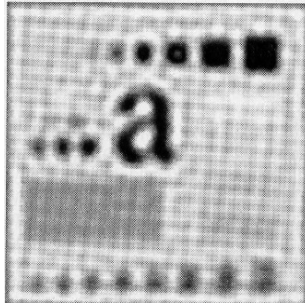
ILPF results



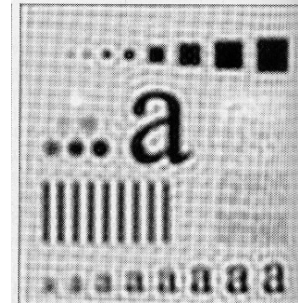
Original image



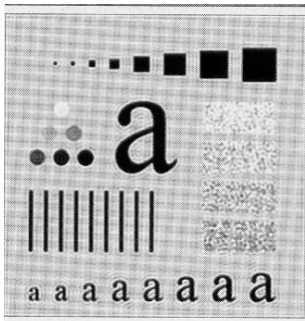
$r_0 = 5$



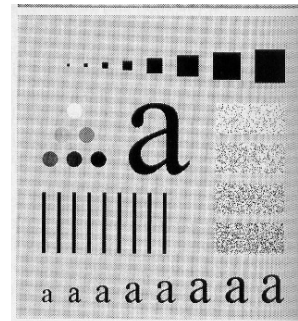
$r_0 = 15$



$r_0 = 30$

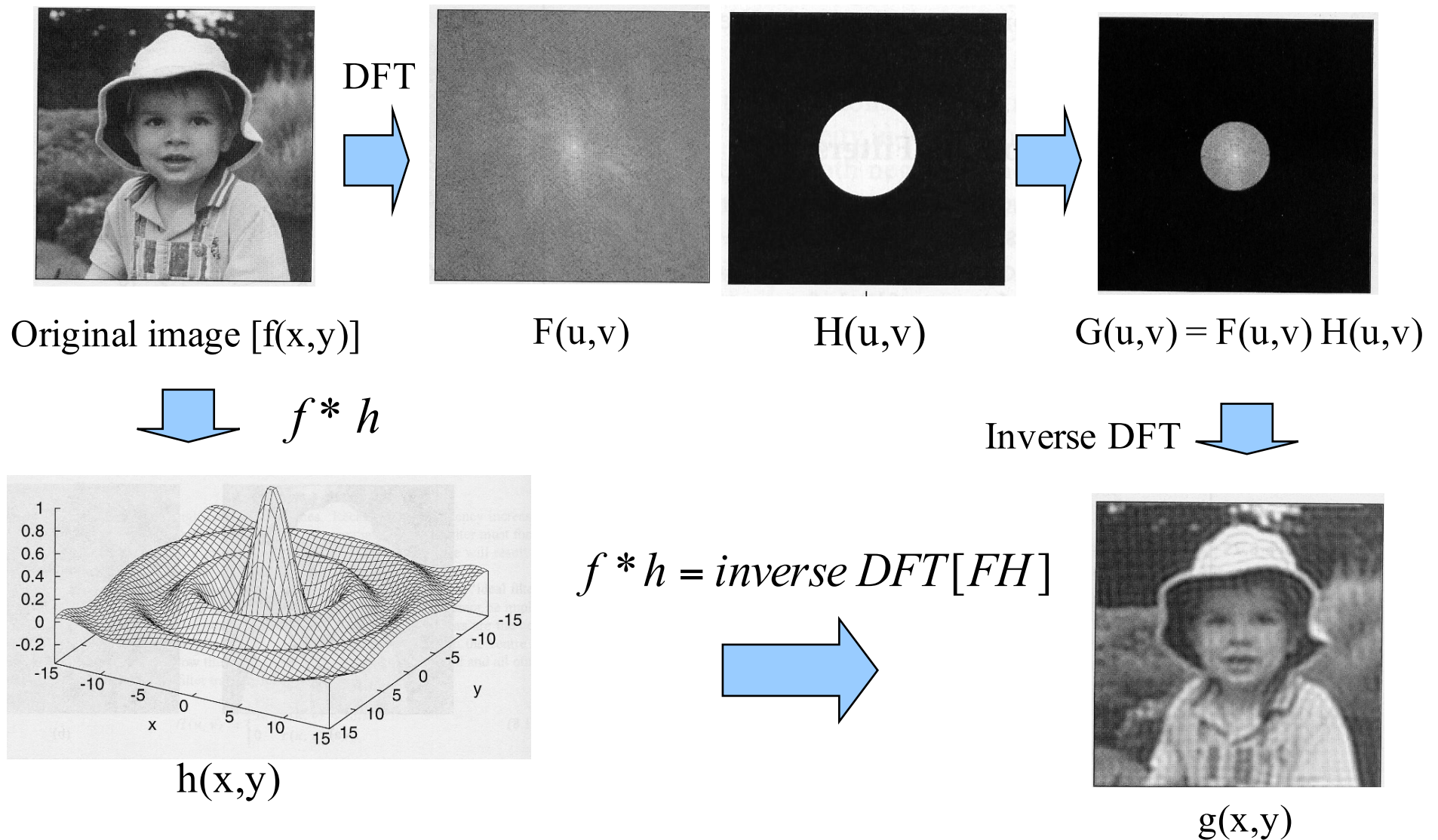


$r_0 = 80$

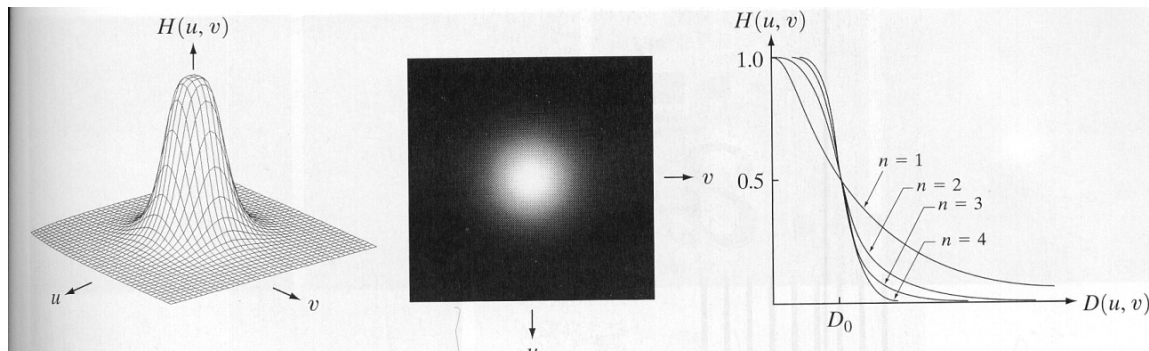


$r_0 = 230$

ILPF ripple effects

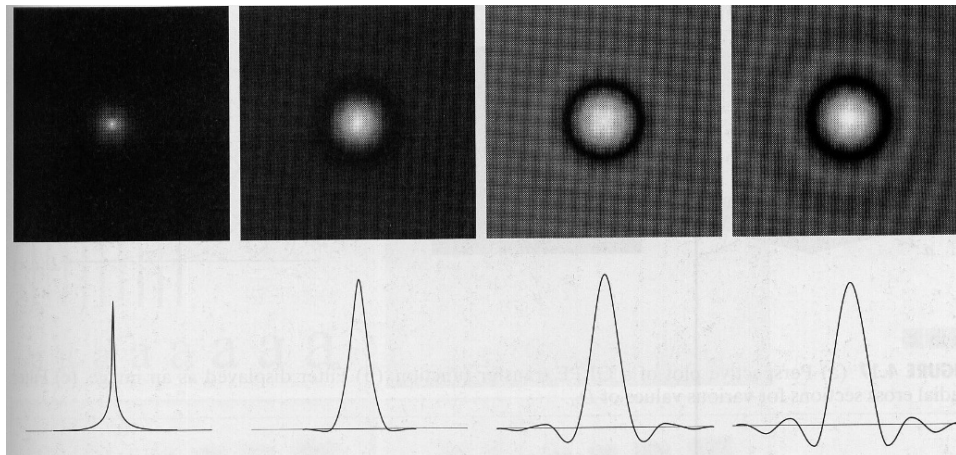


Butterworth Low Pass Filter (BLPF)



$$H(u, v) = \frac{1}{1 + [r(u, v) / r_0]^{2n}}$$

$$r(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$



$h(x, y)$

$n = 1$

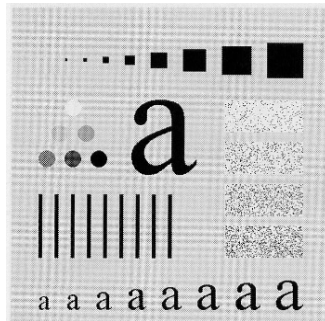
$n = 2$

$n = 3$

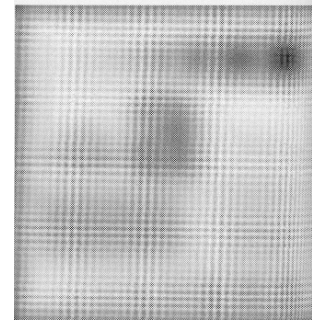
$n = 4$

BLPF results

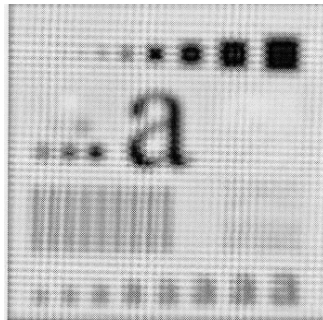
$n = 2$



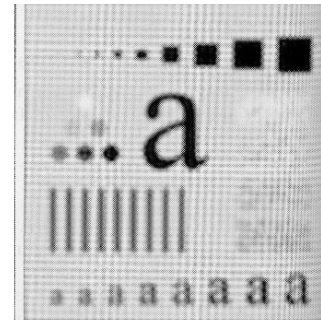
Original image



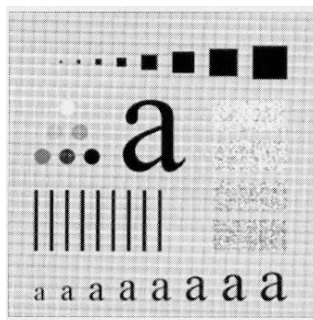
$r_0 = 5$



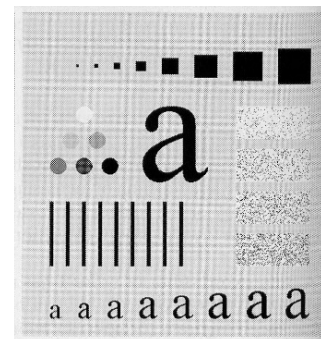
$r_0 = 15$



$r_0 = 30$

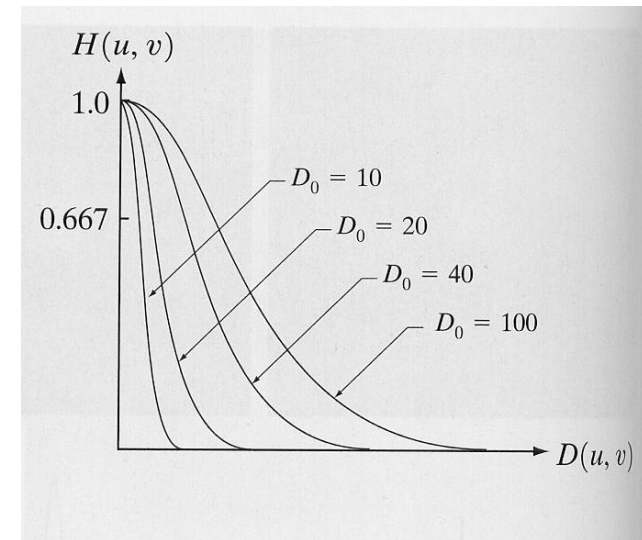
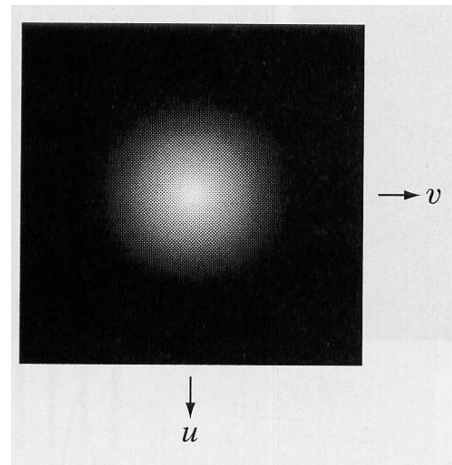
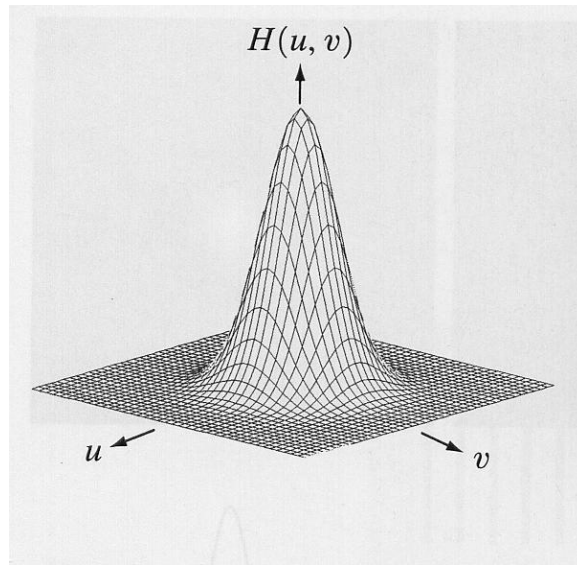


$r_0 = 80$



$r_0 = 230$

Gaussian Low Pass Filter (GLPF)



$$H(u, v) = e^{-r^2(u, v) / 2r_0^2}$$

$$r(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$

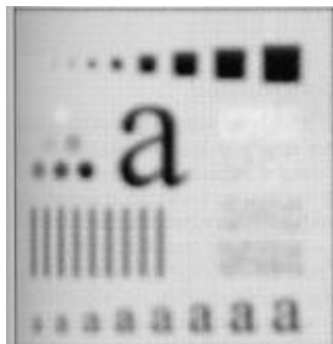
GLPF results



Original image



$r_0 = 5$



$r_0 = 15$



$r_0 = 30$

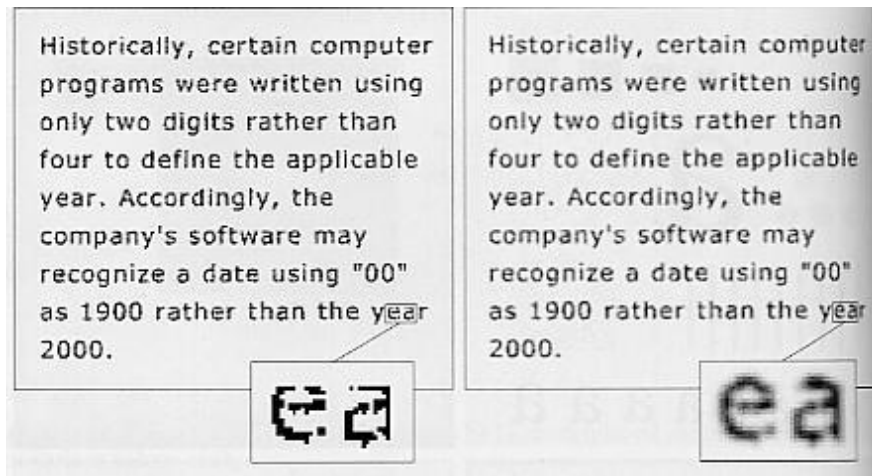


$r_0 = 80$



$r_0 = 230$

Applications of Low Pass Filter

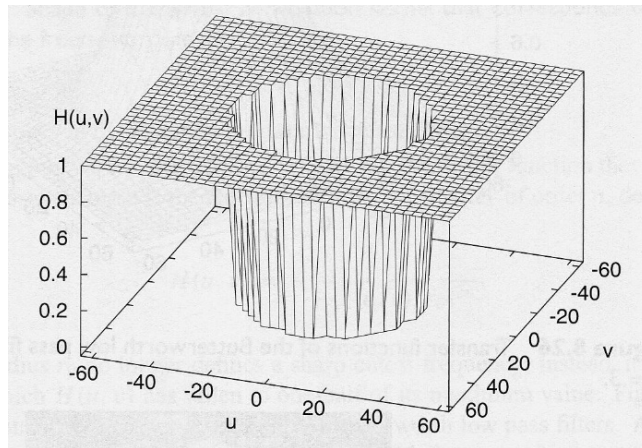


Character recognition



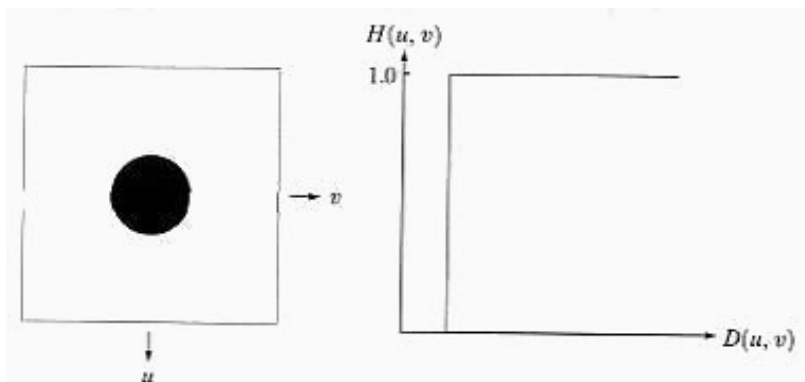
Picture Studio Decoration

Ideal High Pass Filter (IHPF)

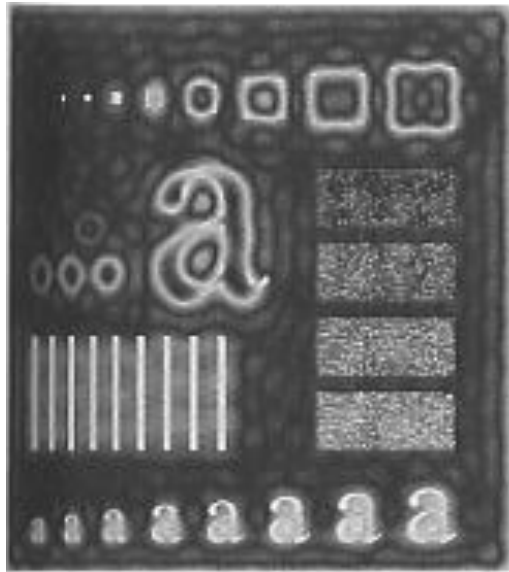


$$H(u, v) = \begin{cases} 0 & r(u, v) \leq r_0 \\ 1 & r(u, v) > r_0 \end{cases}$$

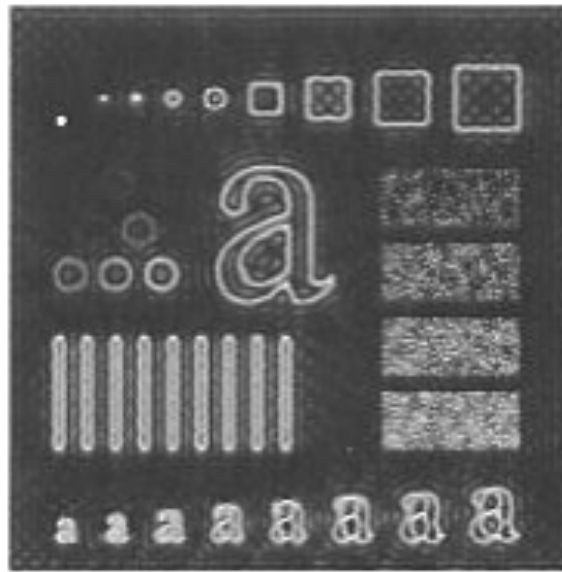
$$r(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$



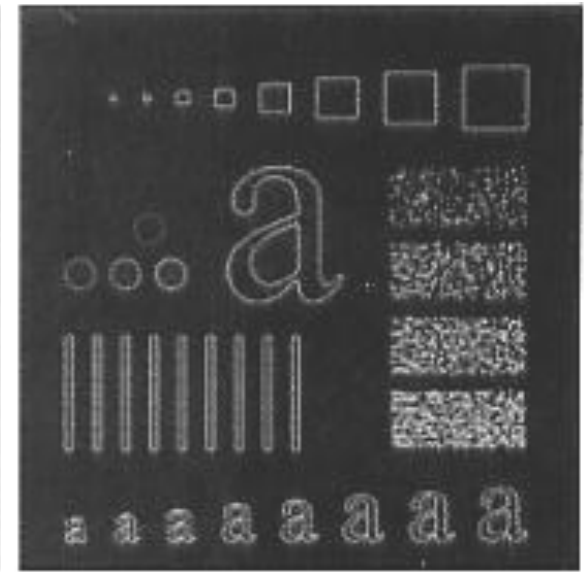
IHPF results



$$r_0 = 15$$

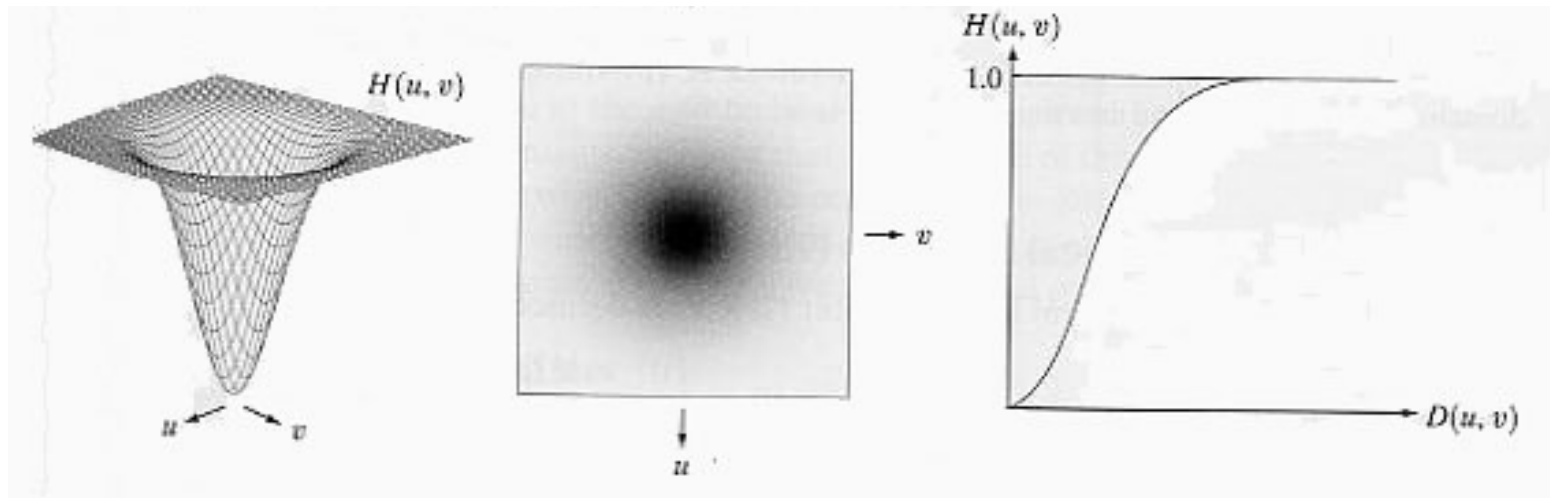


$$r_0 = 30$$



$$r_0 = 80$$

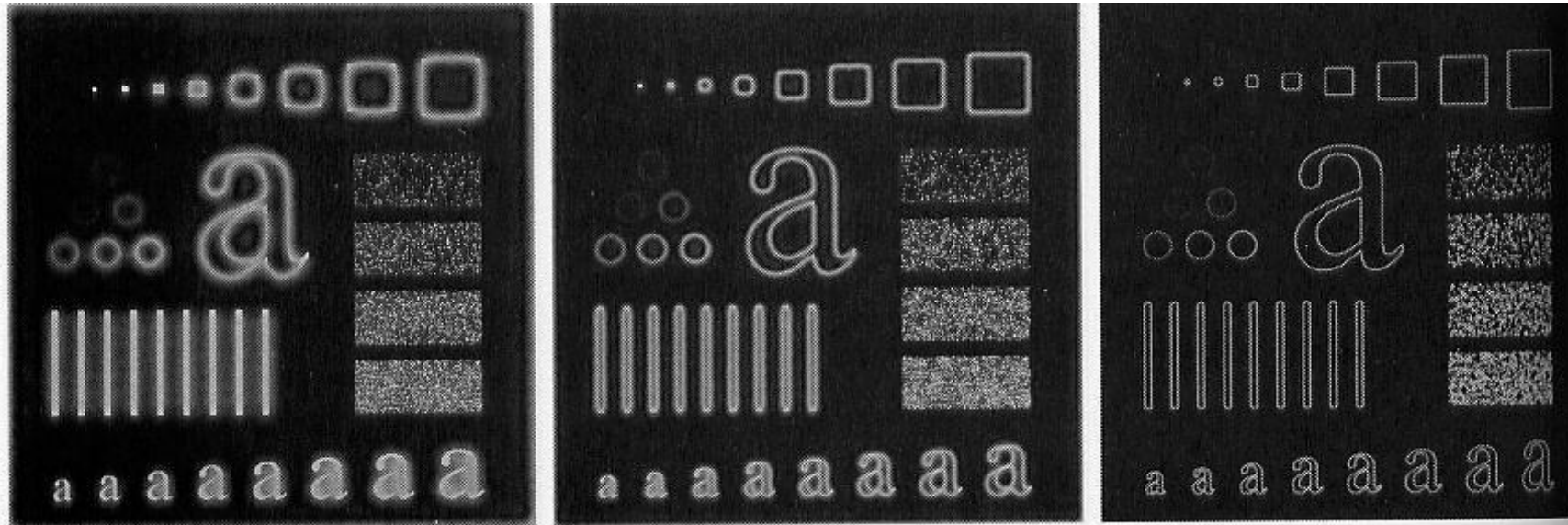
Butterworth High Pass Filter (BHPF)



$$H(u, v) = \frac{1}{1 + [r_0 / r(u, v)]^{2n}}$$

$$r(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$

BHPF results

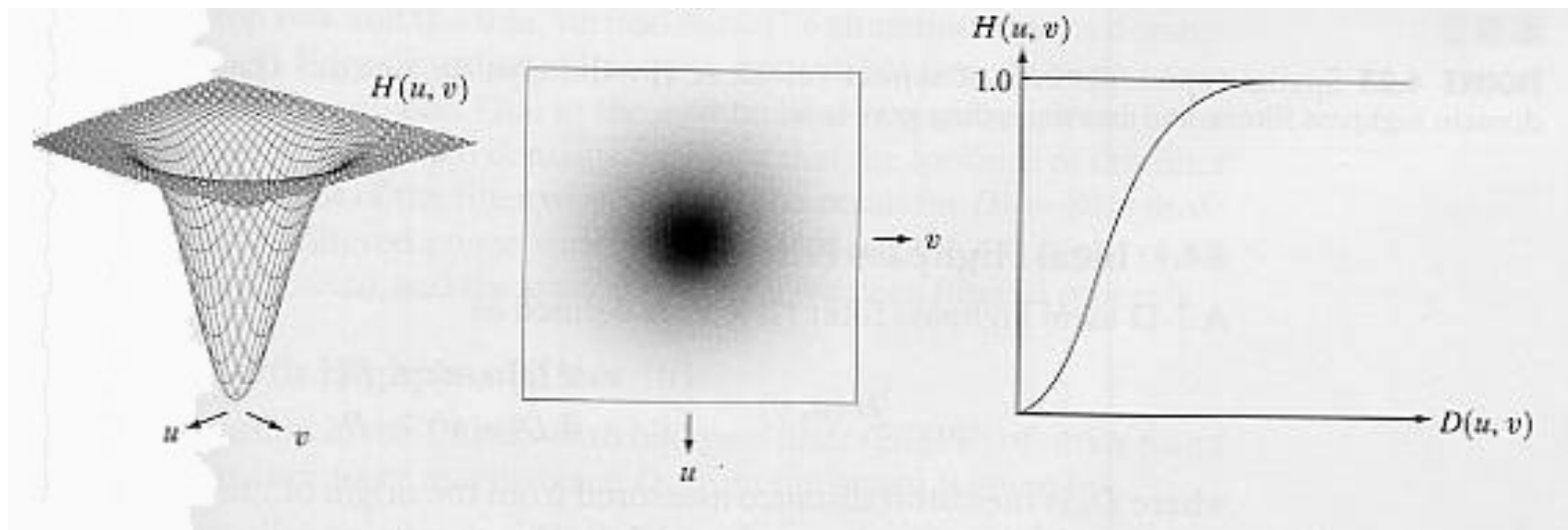


$$r_0 = 15$$

$$r_0 = 30$$

$$r_0 = 80$$

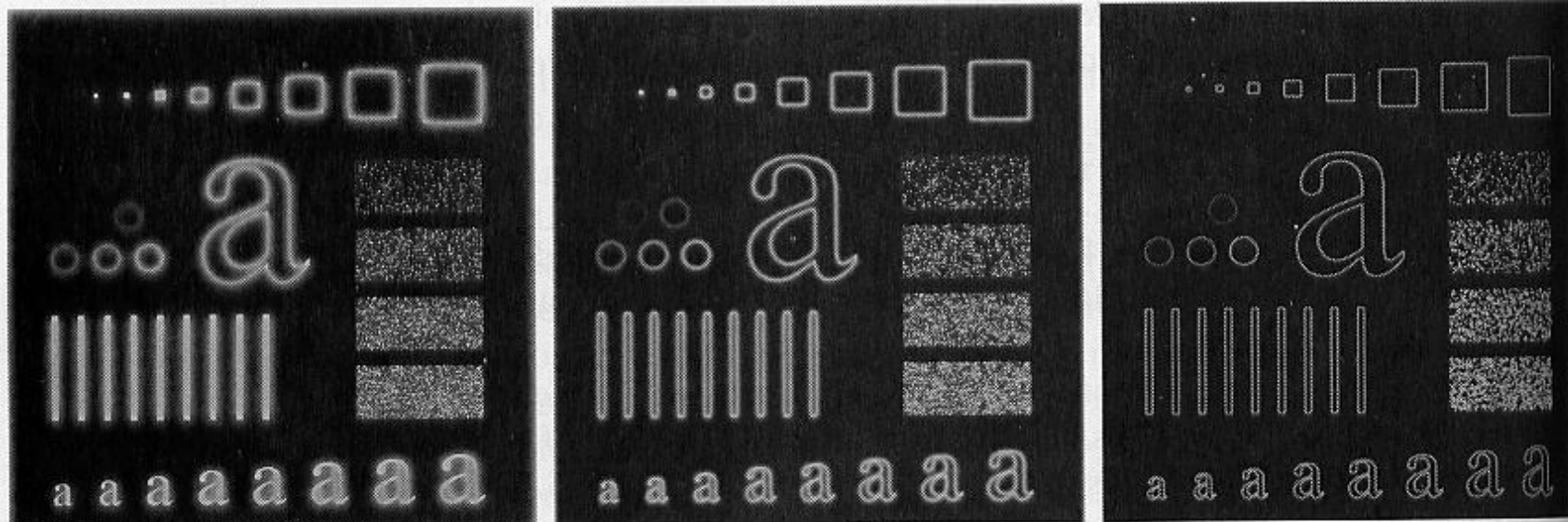
Gaussian High Pass Filter (GHPF)



$$H(u, v) = 1 - e^{-r^2(u, v) / 2r_0^2}$$

$$r(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$

GHPF results

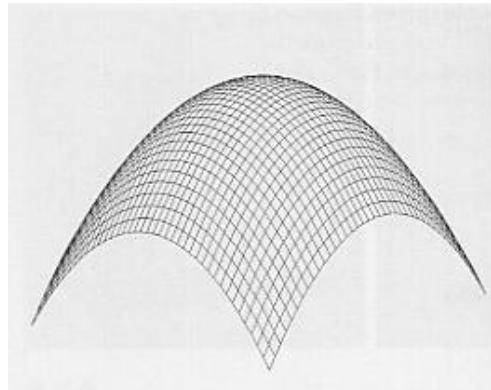


$$r_0 = 15$$

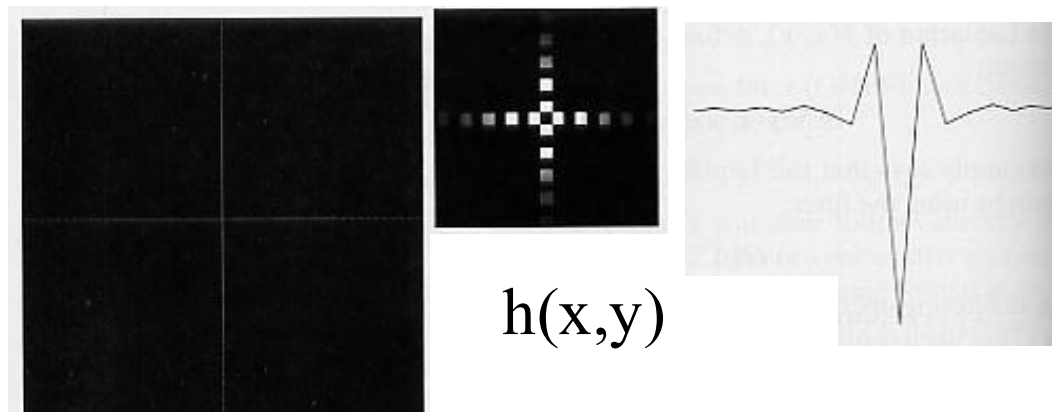
$$r_0 = 30$$

$$r_0 = 80$$

Laplacian Filter (Second-order Filter)



$$H(u, v) = -[(u - M / 2)^2 + (v - N / 2)^2]$$



$h(x,y)$

Laplacian Filter results

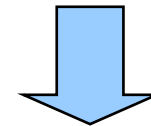


Original image

Laplacian
filter



Laplacian filtered
image



$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

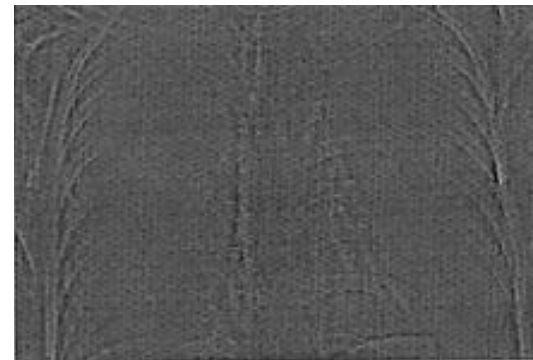


High Frequency Emphasis Filter

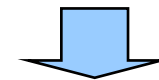


$f(x,y)$

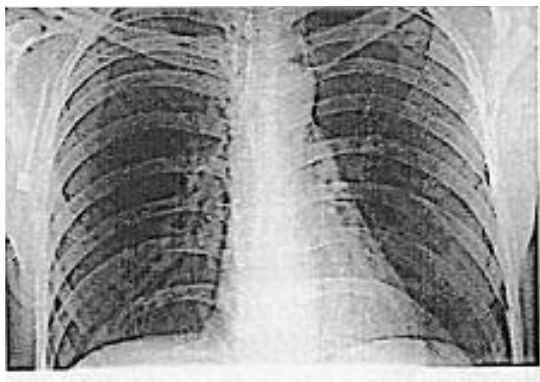
$$H_{hp}(u,v)$$



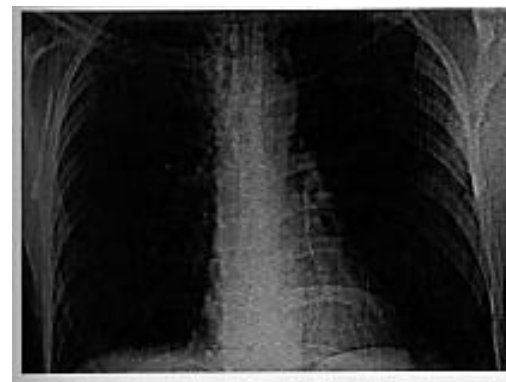
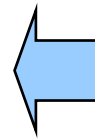
BHPF



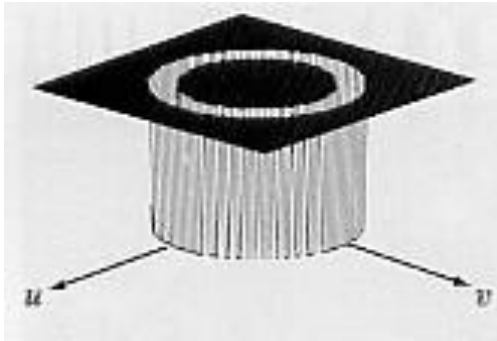
$$H_{hfe}(u,v) = a + bH_{hp}(u,v)$$



Histogram
equalization

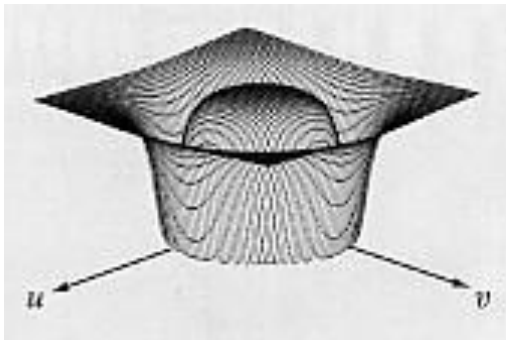


Band Reject Filter (BRF)



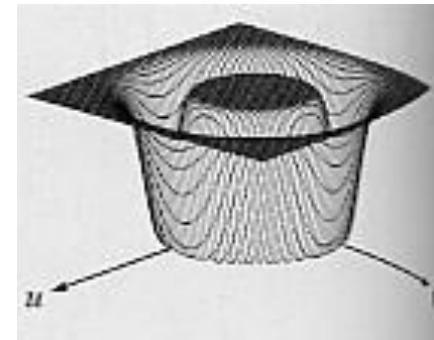
Ideal BRF

$$H(u, v) = \begin{cases} 1 & ; \quad r(u, v) < r_0 - \frac{BW}{2} \\ 0 & ; \quad r_0 - \frac{BW}{2} \leq r(u, v) \leq r_0 + \frac{BW}{2} \\ 1 & ; \quad r(u, v) > r_0 + \frac{BW}{2} \end{cases}$$



Butterworth BRF

$$H(u, v) = \frac{1}{1 + \left[\frac{r(u, v) \cdot BW}{r^2(u, v) - r_0^2} \right]^2}$$



Gaussian BRF

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{r^2(u, v) - r_0^2}{r(u, v) \cdot BW} \right]^2}$$

BRF results

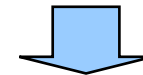


$f(x,y)$

DFT

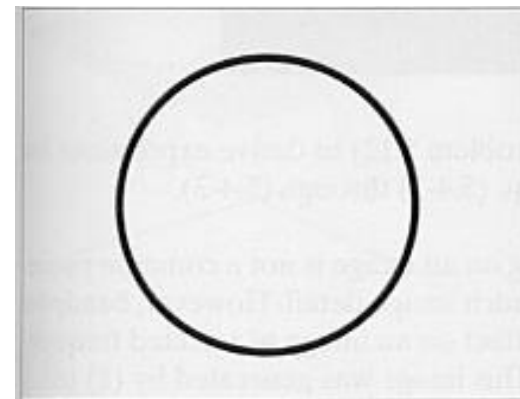
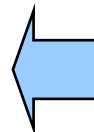


$F(u,v)$



$g(x,y)$

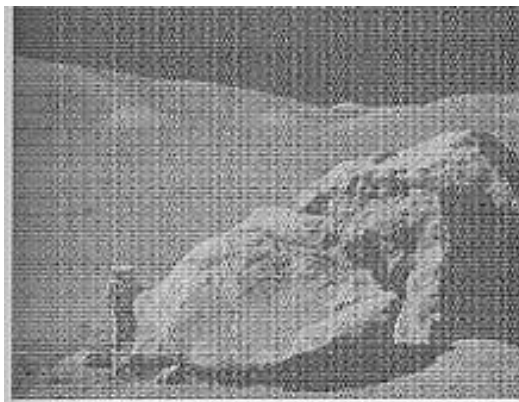
Inverse DFT
 $F(u,v)H(u,v)$



$H(u,v)$

Band Pass Filter (BPF)

$$H_{BP}(u, v) = 1 - H_{BR}(u, v)$$

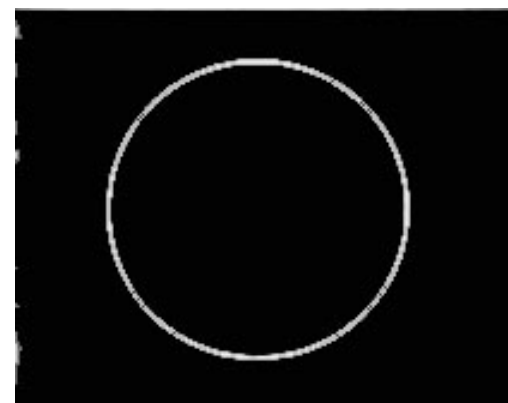


$f(x, y)$

DFT

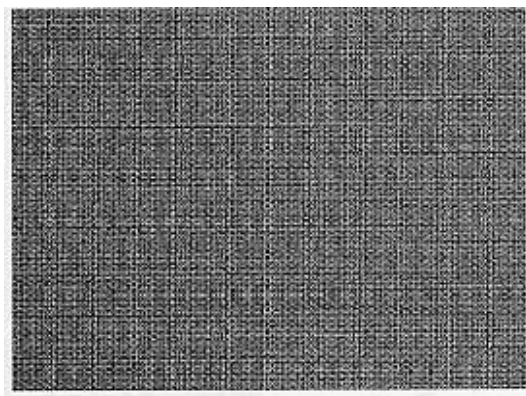
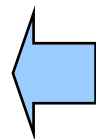


$F(u, v)$



$H(u, v)$

Inverse DFT
 $F(u, v)H(u, v)$



$g(x, y)$